



GÖTEBORGS
UNIVERSITET

DEPARTMENT OF POLITICAL SCIENCE

ANYONE BUT THEM?

Exploring the effect of negative voting on substantive representation in parliamentary democracies

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Master's Thesis:	30 credits
Programme:	Master's Programme in Political Science
Date:	2026-05-26
Supervisor:	Wouter Schakel
Words:	15 956

Abstract

With a growing affective polarization, scholarly attention has turned toward *negative voting* as one of its behavioral expressions. However, as existing research focuses on assessing the prevalence of the phenomenon, particularly in the United States, we know comparatively less about its consequences or dynamics in parliamentary systems. This thesis addresses these gaps by examining to what extent negative voting affects substantive representation in parliamentary democracies, and whether this relationship varies across party systems with different levels of fragmentation. Using post-election survey data from the Comparative Study of Electoral Systems (CSES), covering 164 elections in 44 parliamentary countries between 1996 and 2024, the analysis reveals twofold results. First, there is mixed evidence for a negative relationship between negative voting and substantive representation, with findings remaining sensitive to model specification and operationalization. This implies two things: if there is an effect, it is mainly driven by stable cross-national structural characteristics rather than short-term within-country changes, and that empirical conclusions depend on how the phenomenon is measured. Second, party system fragmentation does not consistently condition the relationship, although additional analysis suggests that the dynamic might differ between two-party systems and multiparty systems more broadly.

Key words: negative voting; affective polarization; substantive representation; congruence; political behavior

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1 Introduction

Against the backdrop of growing affective polarization, contemporary democracies are increasingly shaped by strong emotional hostility toward opposing parties and their supporters. In this context, scholarly attention has turned toward *negative voting* as one of its behavioral expression (Garzia & Ferreira da Silva, 2022a, 2022b; Russo & Areal, 2025; Sulmont & Incio, 2025). This refers to a form of electoral behavior where voters are motivated by antipathy rather than support for a party's policies and platform. In practice, this means that voters select parties or candidates not because of their policy preferences, but to prevent the realization of others (Gant & Davis, 1984; Garzia & Ferreira da Silva, 2022a; Sulmont & Incio, 2025; Weber, 2021).

Despite this growing attention, the existing literature on the phenomenon has mainly focused on assessing its prevalence, giving considerably less attention to its consequences. As a result, we still do not know if negative voting affects substantive representation, despite several explicit and implicit arguments suggesting such a relationship (Gant & Davis, 1984; Garzia & Ferreira da Silva, 2022b; Sulmont & Incio, 2025; Weber, 2021). Elections are commonly understood as the primary mechanisms through which citizens communicate their policy preferences to political elites (Powell, 2004; Stimson et al., 1994). However, if electoral choices are increasingly driven by opposition rather than support, elections may produce less accurate policy signals and thereby weaken the connection between voters and their representatives (Sulmont & Incio, 2025). In turn, as public opinion plays a smaller role in the policy process, this might reduce the ability of elections to ensure both responsiveness and accountability. By extension, exploring whether such an effect exists could help us determine the degree to which negative voting violates a core principle of democracy and should be considered a source of concern.

Additionally, the field of negative voting has this far mainly centered on the United States, despite evidence that negative voting also exists elsewhere (Garzia & Ferreira da Silva, 2022a; Weber, 2021). Consequently, the literature provides limited guidance on how the behavior operates in parliamentary and multiparty systems as many assumptions underlying existing research are closely linked to the logic of binary electoral competition (Garzia & Ferreira da Silva, 2022a). Here, opposition toward one party translates almost automatically into support for the alternative and requires limited strategic reasoning. In more fragmented systems, by contrast, voters face a broader range of parties and strategic trade-offs, potentially altering how negative voting shapes electoral choices and, in turn, substantive representation.

This thesis aims to address these gaps by answering the following two related research questions: First, as the main focus of the thesis: *To what extent does negative voting affect substantive representation between voters and political representatives in parliamentary democracies?* Second, as a more exploratory part: *To what extent does this relationship vary across party systems with different levels of fragmentation?*

Empirically, I draw on post-election survey data from the Comparative Study of Electoral Systems (CSES), covering 164 elections in 44 parliamentary countries between 1996 and 2024. By aggregating individual-level data into repeated country-year observations across a large number of parliamentary democracies, this data provides beneficial opportunities to examine both within-country and cross-national variation in negative voting and substantive representation. Accordingly, by adopting a twofold analytical strategy using both two-way fixed effects models and random intercept and slope models, this thesis provides deeper insight into whether the relationship is driven primarily by changes within countries over time or by more structural cross-national patterns. In addition, by comparing multiple operationalizations of negative voting, the analysis is further able to assess whether the identified relationship depends on how the phenomenon is conceptualized.

In doing so, this thesis offers several broader contributions to our understanding of negative voting. First, it extends the theoretical framework by connecting the growing literature on affective polarization and negative voting to substantive representation and democratic accountability. Second, it broadens the current scope of the literature by exploring its consequences, its existence outside of the United States and whether the assumptions of a two-party system can be translated onto multiparty systems.

The findings of such an analysis are mixed. The main relationship between negative voting and substantive representations is dependent on both how variation in the data is handled and how negative voting is measured, and there is limited support for a conditional effect on party system fragmentation. These findings indicate three things: First, if a relationship exists, it is primarily driven by cross-national differences rather than changes within countries over time. This suggests that negative voting may be embedded in more stable structural or cultural characteristics of political systems, rather than reflecting short-term fluctuations in affective polarization alone. Second, different conceptualizations of the phenomenon lead to different conclusions, making findings across existing studies difficult to compare. Third, the conditional relationship may depend less on party system fragmentation per se and more on broader differences between electoral systems, which warrants further investigation.

Finally, the thesis also highlights broader democratic implications. When political actors mobilize electoral support through antagonism and opposition instead of policy responsiveness, this may create a political climate characterized by symbolic conflicts and weaker accountability. In other words, it might restructure the political landscape in favor of populist and authoritarian forces (Garzia & Ferreira da Silva, 2022b; Pereira, 2019; Sulmont & Incio, 2025).

2 Previous Research and Theory

This section develops the theoretical framework of the thesis in three steps. First, I review existing literature on negative voting, with particular focus on its relevance in contemporary democracies. Second, I examine how and the extent to which different forms of negative voting reflect voters underlying policy preferences. Third, building on this information, I discuss how negative voting is expected to affect substantive representation at the macro-level.

2.1 Evidence of Negative Voting

Although scholarly interest in negative voting has grown in recent years, its existence was already confirmed by research originating in the United States during the 1970s (Gant & Davis, 1984; Gant & Sigelman, 1985; Kernell, 1977). From here on, the behavior is consistently defined in relational terms, emphasizing voting *against* a party or candidate rather than *for* one. More specifically, a negative voter is described as someone who expresses strong dissatisfaction with an opposing party or candidate while remaining neutral toward their own party. In this sense, they are considered to be motivated by antipathy rather than support or policy agreement (Gant & Davis, 1984; Garzia & Ferreira da Silva, 2022b, 2022a; Sulmont & Incio, 2025; Weber, 2021).

Today, attention to negative voting has expanded alongside a growing concern about affective polarization, defined as “the extent to which citizens feel sympathy toward partisan in-groups and antagonism towards partisan out-groups” (Wagner, 2021, p. 1). It is argued that when the political environment becomes increasingly characterized by hostility and perceptions of opponents as threats, it may drive citizens to vote based on instinctive reactions of dislike rather than attachment; thereby, fostering negative voting as a behavioral expression of affective polarization (Garzia & Ferreira da Silva, 2022a; Sulmont & Incio, 2025).

This concern is also empirically confirmed by research on American elections. The relationship between affective polarization and negative voting has, for example, been highlighted in

analyses of Donald Trump's electoral victory over Hillary Clinton in 2016. In this election, the intensity of dislike proved to be a stronger indicator of vote choice than support for one's own candidate (Abramowitz & Webster, 2018; Bankert, 2021). Similarly, using post-election surveys from the 2020 U.S presidential election, Garzia and Ferreira da Silva (2022b) demonstrate that affective evaluations toward political opponents today are the strongest predictor of negative voting.

Moreover, although the literature is heavily centered toward the United States, evidence suggests that similar dynamics are emerging across democratic systems more broadly. Affective polarization is increasingly visible in parliamentary democracies as well (Wagner, 2021) and studies from countries such as Italy demonstrate that voters may support candidates that they otherwise distrust primarily because they oppose a disliked alternative (Garzia, 2014; Garzia & Ferreira da Silva, 2022a; Pereira, 2019).

However, despite this growing concern, only a limited number of studies have explored the political consequences of negative voting. The findings that do exist, however, suggest that negative voting is a mobilizing and meaningful driver of electoral outcomes. Negative voters are just as likely as positive supporters to turn out and participate in elections and the behavior is particularly pronounced in elections involving incumbents, where negative voters disproportionately benefit challengers (Gant & Davis, 1984; Gant & Sigelman, 1985; Sigelman & Gant, 1989). Most notably, combining vote-choice data with party manifestos, Weber (2021) proves that voters respond more strongly to the most distant (i.e., the least liked) policy positions and therefore tend to vote for parties that are more extreme than their own ideological position. He further demonstrates that these micro-level dynamics translate into broader systematic outcomes, as elections with higher levels of negative voting produce significantly more polarized legislatures.

Together, these findings point toward a still unresolved question concerning the relationship between negative voting and democratic representation. As highlighted by Sulmont and Incio (2025), the mere definition of the phenomenon raises broader questions about whether elections continue to function as effective mechanism for translating citizens preferences into policy outcomes. Yet, we still lack a clear understanding of how voting *against* rather than *for* shapes the relationship between citizens and their political representatives.

2.2 Negative voting and Electoral Choice

Against this backdrop, this thesis focuses specifically on exploring the effect of negative voting on *substantive representation*. This refers to the extent to which political representatives act in accordance with the interests and preferences of those they represent and constitutes a core dimension of democratic representation because it concerns what representatives *do*, rather than who they *are* (Dovi, 2018; Golder & Ferland, 2018). Substantive representation is commonly conceptualized along two dimensions: *responsiveness* and *congruence*. While responsiveness captures a dynamic relationship, focusing on how representatives adjust their positions and outputs in response to changes in citizens' preferences, congruence refers to the alignment between citizens and representatives at a given point in time. Accordingly, the two dimensions are closely linked, as responsiveness is generally understood as the mechanism through which congruence is achieved or restored over time (Beyer & Hänni, 2018; Golder & Ferland, 2018).

More specifically, I argue that many of the arguments presented in the literature of negative voting stand in direct tension with conventional models of substantive representation. Underlying these models is the fundamental assumption that citizens vote in line with their preferences and political interests (Powell, 2004; Stimson et al., 1994). In this sense, elections function as the main mechanisms through which citizens communicate their policy demands to political elites so they can represent them. However, if voters select parties primarily to oppose disliked parties, electoral choices may no longer accurately reflect their underlying preferences.

To further assess the extent to which this assumption is challenged, it is therefore first necessary to examine how different forms of negative voting shape the electoral choice. Starting broadly, the literature generally identifies three overlapping components of negative voting (Gant & Sigelman, 1985; Garzia & Ferreira da Silva, 2022a; Kernell, 1977; Sulmont & Incio, 2025; Weber, 2021): two instrumental and rational components that are driven by (1) retrospective performance evaluations and (2) cognitive dissonance and (3) a third affective component driven by negative attitudes toward out-parties and their candidates.

Early contributions to this literature primarily centered around the first two components, both grounded in rational choice theory and in voters' individual evaluations of candidates' and parties' policy positions. The first approach conceptualizes negative voting as a form of retrospective voting in elections involving incumbents. In this literature, a negative voter is typically identified as either a former supporter of the incumbent, or an independent, who evaluates the incumbent's performance negatively and votes against them (Fiorina & Shepsle,

1989; Kernell, 1977). The second approach further includes elements of political psychology, framing negative voting as a rationalization strategy driven by loss aversion and cognitive dissonance. Here, negative voting is described as an expression of a psychological conflict where voters prioritize avoiding the most threatening policy outcome over maximizing overall policy proximity. For example, a negative voter who supports a party's climate policy but strongly opposes its stance on immigration may reject the party entirely, even if its average policy position would otherwise be acceptable (Gant & Sigelman, 1985; Sigelman & Gant, 1989; Weber, 2021).

The third affective component has emerged together with the more recent contributions to literature. As an expression of affective polarization, work situated within this framework conceptualizes negative voting as motivated by negative attitudes toward parties and candidates. In practice, this means that the party choice is driven by the desire to prevent a disliked party or leader from coming to power (Garzia & Ferreira da Silva, 2022a, 2022b; Russo & Areal, 2025; Sulmont & Incio, 2025). A more specific manifestation of these negative affective orientations is *negative partisanship*, which refers to a more stable form of a political identity that is defined primarily through hostility toward opposing parties and their supporters (Iyengar & Wagner, 2025; Russo & Areal, 2025).

The differences between these components become important when considering how closely electoral choices reflect voters' underlying preferences. First, under the retrospective logic, I argue that the considered assumption should remain largely intact. Even when voters negatively cast a ballot against a party, this particular decision remains anchored in evaluations of performance and policy outcomes (Fiorina & Shepsle, 1989; Kernell, 1977). Second, in comparison, the cognitive dissonance framework implies a more substantial departure from voting in line with preferences. Rather than selecting the party closest to their overall position, as assumed in Downs' classic proximity model, negative voting reflects a reversal of this logic (Weber, 2021). The voter prioritizes avoiding a particularly disliked outcome, potentially accepting greater overall ideological distance. As previously stated, empirical research supports this concern, showing that negative voters driven by cognitive dissonance are more likely to support extreme parties than would be predicted by standard proximity (Ibid).

Finally, the affective polarization framework implies the weakest link between voters' preferences and their electoral choices. As highlighted by Sulmont and Incio (2025), because the behavior is driven primarily by opposition to out-groups, it implies a reasoning that can benefit any party or candidate, as long as it is not the party one opposes or parties that cooperate

with it. Under these circumstances, negative evaluations do not have to stem from disagreement over specific policies but may instead be rooted in social identities or generalized out-group hostility. In other words, the party's platform may be secondary or even deemed irrelevant to the electoral decision, increasing the likelihood that voters support parties or candidates whose policies diverge substantially from their own preferences. This is particularly consequential in combination with the evidence suggesting that affective forms of negative voting today are the main drivers of the behavior (Garzia & Ferreira da Silva, 2022b).

Empirical evidence further supports this interpretation. Focusing on negative partisanship specifically, Albert and Costa (2025) illustrate that fear and distrust toward the opposing party influence how individuals perceive the significance, or “stakes,” of that party's electoral victory. As these perceived stakes increase, voters become more willing to trade a less representative candidate from their own party for one that is more likely to win the election.¹

2.3 Hypotheses and Mechanisms

Building on the previous discussion, I further argue that there are implications for a relationship between negative voting and substantive representation at the macro-level. Specifically, I theorize that negative voting undermines substantive representation by disrupting what Stimson et al. (1994) call *electoral connection* and *rational anticipation*. These are two separate mechanisms that together create a dynamic system where public opinion translates into political decisions and thereby representation.

Starting with the first mechanism, electoral connection achieves representation through electoral punishment. In this model, citizens act as “effective agents” by voting for the parties or candidates that share their policy preferences at a given time (Stimson et al., 1994, p. 30). This initiates a causal sequence where the elected officials make decisions that mirror the opinion of the electorate. If they do not, or if there are changes in public opinion, citizens evaluate their behavior and replace officials with new representatives whose position better reflects these changes in the next election.

I argue that negative voting could undermine electoral connection by disrupting its whole causal sequence. As discussed in the previous section, the behavior weakens the assumption that

¹ Note, however, that their design focuses on partisans (positive and negative) and thus captures trade-offs among voters with baseline in-party attachment and an existing platform against which candidates can be evaluated. Negative voters represent an extreme case of the logic identified by the authors. They are voters for who the antipathy is not counterbalanced by in-party affection and for who the willingness to sacrifice representation has already translated into actual voting behavior.

citizens are effective agents since their electoral choices might no longer represent underlying policy preferences. In turn, this should also make it increasingly difficult for parties to adopt policies that correctly mirror public opinion, as well as affect the ability of elections to correct representational imbalances over time. Shifts in citizens preferences *should* generate electoral punishment and replace representatives whose positions no longer correspond to public opinion. However, when voting decisions are driven by opposition and avoidance, voters may continue to support parties that do not reflect their preferences simply because they remain preferable to the disliked alternative (Sulmont & Incio, 2025).

In contrast, according to the second mechanism of rational anticipation, representation is achieved through the actions of political elites and their tactical adaptation to avoid being electorally replaced. In this model, the agents are described as rational politicians who are committed to their careers and therefore anticipate future electoral reactions and consequences. Instead of waiting to be replaced, these elites adjust their behavior in response to changes in public opinion (Stimson et al., 1994).

However, when elections are increasingly driven by negative voting, I argue that it may undermine rational anticipation in two related ways. First, it may directly undermine the political elites actual ability to anticipate changes in public opinion, as they receive less reliable information about what voters want. Second, because the electoral connection is also undermined, making them less constrained by electoral punishment and accountability, they may also face weaker incentives to adhere to specific policy commitments. As pinpointed by Sulmont and Incio (2025), when supported by negative voters the representatives fulfill the mandate simply by being in opposition and their electoral success does no longer depend on policy approval.

In summary, negative voting is expected to undermine two levels of substantive representation, the process where voters act by replacing political elites and the strategic adjustment of the elites themselves to avoid being replaced. Compared to a situation in which voting primarily reflects policy proximity and positive attachment, higher levels of negative voting should therefore be associated with lower levels of substantive representation between citizens and political representatives (Stimson et al., 1994; Sulmont & Incio, 2025).

H1: Higher levels of negative voting lead to lower levels of substantive representation.

Beyond this general hypothesis, I believe that there is also good reason to expect that the effect of negative voting on substantive representation varies across institutional contexts. However,

as research on negative voting remains heavily centered on the American two-party context, we know comparably little about how negative voting operates in parliamentary and multiparty systems.

I argue that this gap is theoretically important because the behavioral logic underlying negative voting may operate differently depending on the degree of party system fragmentation. In other words, I argue that the extent to which negative voting undermines substantive representation is likely to depend on the number of viable electoral alternatives available to voters (i.e., party system fragmentation).² In systems with lower levels of fragmentation electoral choice is more limited and concentrated among a smaller number of viable alternatives (Garzia & Ferreira da Silva, 2022b). In a two-party system for example, such as the United States, strong aversion toward one candidate or party, therefore, translates almost mechanically into support for the alternative (if voters do not abstain, a behavior that existing research shows negative voters are no more likely to adopt than positive voters). As a result, negative voting in systems with few effective parties requires relatively little strategic reasoning.

As party system fragmentation increases, however, negative voting should operate under more complex conditions. Voters who reject a particular party are not presented with a single default alternative but instead face a broader set of parties. Negative voters must therefore navigate multiple dimensions of choice, including party size, possible party groups and strategic trade-offs. For example, if a negative voter seeks to secure an electoral victory in opposition to the party to which the antipathy is directed, they must vote for a party they know will stand outside such a coalition. It is also likely that strong aversion targets multiple parties, further complicating the electoral choice.

Turning to previous research on substantive representation, a common exception is that party systems with a larger and more ideologically diverse set of parties should produce better representation. Even though the results are mixed, this argument is typically justified by the fact that a broader range of parties allows the ideological distribution of parliament to more closely mirror the distribution of preferences within the electorate. Conversely, systems with two (or a few) parties tend to induce parties to converge toward the median voter, limiting representational diversity (Golder & Ferland, 2018; Golder & Stramski, 2010; Soroka & Wlezien, 2015).

² This definition is borrowed from Valentim and Dinas (2024) who studies the effect of party system fragmentation on the quality of democracy.

However, when considering how the effect of negative voting might vary depending on the extent of party system fragmentation, the implications are less straightforward. On the one hand, the less automatic nature of choice in multiparty systems may urge voters to engage in more deliberate and evaluative reasoning. With a broader set of viable alternatives, negative voters are forced to weigh trade-offs and make more considerate choices. This might make it easier to select a party that both opposes the disliked actor and remains relatively close to their own policy preferences, mitigating the loss of representation.

On the other hand, the increased complexity of multiparty competition could instead intensify a negative effect on representation. If a disliked party is expected to participate in a governing bloc, negative voters may feel compelled to abandon otherwise proximate parties and instead support more distant alternatives that are clearly excluded from such coalitions. While negative voting might also induce bloc shifting in systems with lower fragmentation, such shifts typically involve fewer available alternatives and should therefore limit the ideological movement. As noted above, two-party competition tends to pull both parties toward the center of the voter distribution (Golder & Ferland, 2018; Golder & Stramski, 2010; Soroka & Wlezien, 2015). In multiparty systems, by contrast, strategic avoidance of a disliked party may require larger ideological departures, increasing the risk that negative voting produces choices that deviate from proximity.

In summary, while party system fragmentation is expected to condition the relationship between negative voting and substantive representation, the theoretical framework does not provide a clear expectation regarding the direction of the effect. This second part of the analysis will thus be explanatory.

3 Research Design

This section is structured into three main parts. First, I present the data used in the analysis and the restriction to parliamentary system. Second, I introduce the main variables for this thesis, including the operationalization of negative voting and congruence, as well as the effective number of parties as an interaction term and a set of control variables. Third, and last, I outline the empirical strategy based on two separate model specifications and discuss the assumptions underlying these models.

3.1 Data and Operationalizations

To empirically test my main hypothesis and more exploratory question, I will rely on individual- and country-level data from the ongoing *Comparative Study of Electoral Systems* (CSES), where participating countries administer a common module of survey questions in their post-election studies. The resulting data consists of voting, demographic, district and macro-level electoral system variables that are merged into a single dataset, standardized for comparative and cross-country analysis. Importantly, this ensures that there is the necessary information for me to assess the main variables in this thesis, such as voter evaluations of parties, self-placements on the left-right scale and party positions. Additionally, since the analysis of this thesis is conducted at the country-year level, all individual level data is aggregated accordingly.

To further ensure comparability across institutional contexts and conceptual consistency in measuring substantive representation, the analysis will be restricted to parliamentary systems. As highlighted by Golder and Stramski (2010), the legislature in such countries is the main arena for political power and the distribution of seats therefore provides a clear connection between voter preferences and representation. This makes it possible to conceptualize substantive representation among representatives as a single distribution that can be directly compared to the distribution of the electorate. In contrast, presidential systems divide political authority between the executive and the legislature, making representation the outcome of multiple institutional processes and less straightforward to capture as a single representative distribution. Accordingly, the remaining sample includes over 19 613 individual-level observations across 164 elections in 44 countries, with voter evaluations of over 590 parties. With selected variables from Modules 1-6, the observations span between 1996 and 2024.

Negative voting

While the CSES data limits the measurement of negative voting to an indirect approach, its cross-national and longitudinal coverage is essential for testing the effect on substantive representation in parliamentary systems and across systems with different levels of party fragmentation. This reflects a central trade-off in the research design; I rely on voters' affective evaluations of parties as a proxy for underlying vote motivations, instead of directly observed behavior. As a result, an indirect measurement of negative voting may either capture broader attitudinal patterns that do not always translate into a negative vote (false positives) or risk excluding actual negative voters due to arbitrary thresholds and restrictions (false negatives). An alternative would be to use direct measures of negative voting (see e.g., Gant & Sigelman,

1985; Garzia & Ferreira da Silva, 2022b), which would require survey questions that explicitly ask respondents whether their vote was primarily motivated by support for one option or opposition to another. While such measures offer higher validity, they are typically confined to specific countries and elections, thereby limiting both temporal variation and cross-national comparability. By moving beyond a single-country focus, the CSES offers greater external validity and allows the findings to be generalized to a broader set of institutional settings, thereby addressing the dominance of the U.S.-centered focus in the existing literature.

Although several studies on negative voting have applied indirect approaches, there is no consensus on a standard measurement and the concept continues to be operationalized in diverse ways. As a result, it becomes difficult to separate whether the empirical findings reflect the effects of negative voting itself or are driven by differences in how the concept is measured. For instance, Gant and Davis (1984) rely on open-ended “like/dislike” survey questions to classify respondents as negative voters based on the absence of positive motivations. In contrast, Kernell (1977) captures a more aggregate and behavioral dimension by examining asymmetries in the effects of presidential approval and disapproval on electoral outcomes, classifying negative voting when disapproval has a stronger impact than approval. Even though both approaches rely on survey data from the United States, it is unclear whether they identify the same underlying phenomenon or distinct processes.

To address this gap, I develop and test multiple operationalizations of negative voting to gain a more systematic assessment of whether the observed relationships are consistent across different measures. In doing so, this thesis also aims to contribute to a more transparent and potentially reusable way of measuring negative voting in comparative research. Accordingly, in what follows, I develop three different approaches based on the findings of previous literature, each reflecting a different position along the trade-off between false negatives and false positives.

Operationalization 1: Affective thresholds

The first operationalization is built on the results from Garzia and Ferreira da Silva (2022b), who, based on a direct classification of negative voters, analyze their evaluations of parties using scores on a feeling thermometer (p. 306).³ By doing so, the authors provide us with important guidance on how an indirect measurement based on affective party evaluations can

³ Garzia and Ferreira da Silva (2022b, p.306) used the following question to classify whether a respondent is a negative voter: “Would you say your vote is more a vote for Trump [Biden] or more a vote against Biden [Trump]?”

serve as a proxy for negative voting. More specifically, the authors show that negative voters, as expected by the definition, express almost neutral feelings for their chosen party or candidate and strong negative feelings toward the opponent. However, they also show that positive voters, in comparison, exhibit both relatively high levels of in-group love *and* out-group hate. This suggests that negative voters are not primarily distinguished by stronger hostility toward out-group parties. Instead, positive and negative voters are relatively similar in their dislike of opponents and the main difference lies in the level of affection toward their own party (Ibid).

Using this information as a starting point, I classify a negative voter as an individual who (i) is neutral toward the party they voted for and (ii) express a strong dislike toward another party/parties. To identify this, I use the CSES like-dislike scale where respondents rate the major parties in their country (A-I) on a scale from 0 to 10, where 0 indicates strong dislike and 10 indicates strong like. These party evaluations are then linked to each respondent's reported vote choice in the current election, allowing me to identify both the affective evaluation of their own party and the most strongly disliked party/parties.⁴ A respondent is then classified as a negative voter when they evaluate their chosen party below 7 (5 represents the midpoint of the scale and thus a neutral evaluation) and express strong dislike toward at least one other party by rating them below 3. Empirically, the measure identifies a moderate share of respondents as negative voters in the total sample (26.5 %).

However, while this measurement mirrors the conceptual definition and previous findings of negative voting, the use of arbitrary thresholds might generate both false positives and negatives. I further argue that there is an increased risk of including false positives when the analysis is set in parliamentary systems with more parties. This operationalization identifies negative voters by comparing affect toward the in-party and the out-party, which maps directly onto a binary electoral choice where the parties are each other's challengers by default. In such a context, it might be reasonable to assume that the feelings toward Party B are what actually motivated the choice of Party A, for which the respondent is neutral. However, this assumption becomes less straightforward in multiparty systems where voters face several alternatives. For example, a voter may support a smaller party to enable a certain government coalition or to keep them in parliament, while simultaneously rating another party negatively. Such a case could meet the criterion above without the vote necessarily being an expression of antipathy. Similarly, the measurement may capture voters who hold generally negative views of all parties

⁴ Note that the sample is restricted to respondents who report an actual vote choice.

and simply choose the “least bad” option, rather than actively voting against a specific opponent.

Consequently, the following two operationalizations will be constructed with the aim of better capturing the phenomenon across party systems with more parties.

Operationalization 2: Relative affective evaluations

Building on the framework of Garzia and Ferreira da Silva (2022b) once again, the second operationalization adopts a relative, within-respondent approach by standardizing party evaluations on the like-dislike scale. To do this, I use the math formula for the z standardization. Unlike the previous operationalization, this measurement accounts for the full set of available parties rather than relying on a binary comparison between a chosen party and a single opponent. Also, instead of classifying voters based on absolute values, it captures how each party is rated relative to the respondent's own ratings. This reduces sensitivity to arbitrary cut-offs and improves comparability by accounting for individual differences in the usage of the scale (e.g., generally generous or critical raters). Accordingly, for each respondent i and party p , I compute a standardized score as follows:

$$z_{i,p} = \frac{X_{i,p} - \bar{X}_i}{SD_i}$$

Where $z_{i,p}$ represents the standardized score (z-value) for respondent i and party p , $X_{i,p}$ is the rating that respondent i gives to party p on the 0–10 dislike–like scale, $\bar{X}_i$ is the mean of all ratings provided by respondent i across all parties, and SD_i is the standard deviation of respondent i 's ratings across all parties. To capture the characteristics of strong dislike toward the out-group and neutral feelings toward the in-group, a respondent is then classified as a negative voter if the most disliked party receives a low relative evaluation ($z \leq -1$), indicating at least one standard deviation below the respondent's mean, and the chosen party is evaluated at most weakly positive ($z \leq 0.5$).

An important consequence of this operationalization is that it applies stricter classification criteria than the first measurement, thereby reducing the risk of false positives by excluding respondents whose party evaluations are only moderately differentiated. It also avoids attributing negative motivation to cases where voters are generally negative against all parties. At the same time, this stricter criterion increases the risk of false negatives, as some negative voters may not display sufficiently extreme differences in their evaluations. This is also reflected empirically as the measure identifies a small share of voters in the sample (5 percent).

Operationalization 3: Ideological deviation

The third operationalization adds a component of ideological distance to the first threshold measure. This is done with the aim to shift the focus from affective evaluations alone to the spatial relationship between voters and parties, drawing on Weber's theory of the "tragedy of negativity" (2021, p.2). As previously stated, his paper shows that negative voters deviate from proximity-based voting and select parties that are *not* the closest option on the left-right scale. For this, I use the CSES left-right scale, where respondents place themselves and political parties on a common 0-10 scale, with lower values indicating left-wing positions and higher values indicating right-wing positions. For each respondent, I first calculate the ideological distance between their self-placement on the scale and the mean position of each party. I then identify the party that is closest to the respondent in ideological terms. To further ensure that this deviation reflects negative motivation rather than random or uninformed choice, I combine this condition with the previous affective thresholds. A respondent is coded as a negative voter if they (i) do not vote for the ideologically closest party, (ii) is neutral toward the party they voted for and (iii) express a strong dislike toward another party/parties.⁵

This operationalization offers some potential advantages in comparison to previous measures. First, by combining observed deviations with affective evaluations, the operationalization can identify situations where voters abandon their closest ideological position in favor of another party for which they express weak or neutral feelings. This makes it more consistent with the theoretical expectation of negative voting as driven primarily by opposition rather than positive attachment. Second, by considering the ideological relationship between voters and all relevant parties in the system, the measure remains applicable in multiparty contexts where voters face several viable alternatives rather than a single default opposition party. For example, it helps us distinguish negative voters from respondents who express weak partisan attachment in general and who nevertheless vote for their closest party and remain consistent with proximity voting.

This measure identifies a somewhat smaller share of respondents as negative voters than the threshold measure alone (23 percent). However, while this additional requirement may strengthen the conceptual validity of the measurement, it does not solve all potential issues

⁵ Note that Weber (2021) argues that there is a centrifugal dynamic of cognitive dissonance. He writes that while negative voters support more ideologically extreme parties than their closest alternative, they remain on the same side on the left-right scale. I choose to adopt a broader understanding of ideological deviation. In contexts with higher levels of affective polarization, I argue that voters also may abandon their closest ideological option in favor of more moderate or even cross-bloc alternatives if these are perceived as more effective opponents to a disliked party. Therefore, the operationalization does not require the chosen party to be more ideologically extreme than the closest alternative.

associated with the first threshold measurement. As with all indirect measurements, it remains difficult to determine if antipathy is what actually motivated the party choice. Deviations from proximity may still arise from alternative behavior unrelated to negative voting, such as strategic voting or limited political information. Moreover, because the operationalization relies on self-reported ideological placements and affective thresholds, it remains sensitive both to individual perceptions of the left-right scale and to how respondents interpret and use evaluative scales more broadly. In summary, this suggests that the operationalization still includes risks of both false positives and negatives.

(In)Congruence

To measure the effect of negative voting on substantive representation, I will specifically focus on the level of *congruence* between the electorate and its representatives rather than responsiveness. This choice is motivated in two ways. First, although negative voting is expected to affect both parts of substantive representation, the effect on responsiveness is likely to be reflected in congruence. If negative voting weakens electoral connection or alters how representatives anticipate voters' preferences, this should ultimately manifest as a reduced alignment between citizens' preferences and policy outcomes (Golder and Ferland, 2018). Second, congruence can be measured more consistently across countries using the left-right scale.

The operationalization of congruence is built on what Golder and Stramski (2010) call *many-to-many* congruence, measured as the extent to which the ideological distribution of the legislature mirrors the distribution of preferences in the electorate. This approach follows the author's argument that ideological congruence should be conceptualized differently depending on whether the relationship of interest links *one* citizen to *one* representative or *many* citizens to *many* representatives. Before this contribution, congruence was commonly calculated as the difference between the mean ideological positions of citizens and the elites (e.g., Powell, 2004). However, by treating distributions as equivalent in size and dispersion, we lose information about differences between sample variances and fail to distinguish between a more centrist electorate and one that is highly polarized (Lupu et al., 2017; Schakel & Hakhverdian, 2018). The many-to-many approach is designed to address these limitations by comparing the full distributions of citizens and representatives rather than averages, capturing both the spread and shape of preferences (Golder & Ferland, 2018; Golder & Stramski, 2010; Lupu et al., 2017).

Empirically, the many-to-many relationship has been measured in several ways. The two most common calculations are the degree of overlap between cumulative distributions (e.g., Golder and Stramski, 2010) or probability distributions (e.g., Lupu and Warner, 2017). While these approaches provide an intuitive summary of how similar the distributions are, they also lead to loss of information. Such overlap-based measures require researchers to make prior decisions about how to define this overlap, which in practice involves grouping the data into sets of bins or histograms. This process eliminates within-group variation and compresses potentially meaningful differences between observations. Additionally, these measures ignore all observations in areas without common support, that is, portions of the distributions that do not overlap, thereby excluding additional and meaningful information. As a result, the estimated level of congruence can be sensitive to arbitrary choices in how these bins are constructed, allowing different levels of overlap for the same underlying distribution (Lupu et al., 2017).

To avoid these limitations, I use the Earth Mover's Distance (EMD) to measure congruence following Lupu et al. (2017). EMD was originally developed in computer science and calculates the minimum amount of "work" (calculated as mass multiplied by distance) needed to transform two distributions so that they are identical. Applied to congruence, this can be interpreted as how much of the distribution of representative positions would need to be moved, and how far, to match the distribution of citizens' positions. In simple terms, the greater the amount of "mass" that must be moved over a longer distance, the lower the level of congruence. Compared to earlier approaches, EMD operates directly on the full distributions instead of relying on predefined overlap measures, allowing it to capture differences across the entire range of the ideological spectrum. Moreover, because the EMD is based on all pairwise distances between points in the distribution, it incorporates both the magnitude and location of the differences (Ibid). For my analysis, this means that the measurement assesses both *how much* citizens and representatives differ and *where* these differences occur, providing a more meaningful assessment of congruence.

In practical terms, I operationalize congruence by applying the EMD to left-right distributions constructed from the CSES data. Specifically, I use the left-right scale and derive the distribution of citizens from the respondents' self-placement, while the party positions are approximated by the mean of voter placements for each party, weighted by its seat share in the lower house. Congruence is then measured as the EMD between these two distributions. Given the 0-10 scale, the resulting values can be interpreted directly as the average ideological distance required to align the representative distribution with the electorates, where lower

values indicate greater congruence and higher values indicate lower congruence (Ibid). In the empirical analysis, the dependent variable will therefore be referred to as *incongruence*.⁶

Nevertheless, the choice to rely on the left-right scale also involves important limitations. First, the interpretation of the scale may vary across countries and time, which introduces challenges for comparability (Lindqvist et al., 2025). For example, while it is commonly used in the literature on substantive representation as a general ideological dimension (e.g., Golder & Stramski, 2010), its substantive meaning is not fixed and may increasingly incorporate additional dimensions, such as GAL-TAN, in later survey waves. As a result, the scale may capture different underlying conflicts both across and within contexts, even when the numerical values appear comparable.

Second, while some studies suggest that the left-right scale produces similar patterns to those based on more specific policy issues (Lupu & Warner, 2022a), it reduces political competition to a single summary index, thereby overlooking potential (mis)matches between citizens and representatives (Lupu et al., 2017). A more comprehensive measure of congruence would instead rely on multidimensional policy preferences, allowing for a better assessment of substantive representation across several issue areas (Schakel & Hakhverdian, 2018). This would also better leverage the strengths of the EMD measure, as distances could be calculated across multiple policy dimensions rather than along a single dimensional scale (Lupu & Warner, 2022b). However, implementing this type of measure requires detailed and comparable cross-national data on issue positions for both voters and representatives, which is not available in the CSES. As a result, adopting a multidimensional approach would reduce the number of cases and undermine the cross-national aim of the analysis.

Another potential concern relates to the use of the left-right scale in both the dependent variable and the independent variable, which is the case for the ideological deviation operationalization of negative voting. While they capture distinct phenomena, individual negative voting behavior and the gap between the overall distribution of voters and representatives, they rely on a common ideological dimension. This introduces a risk of mechanical correlation, as a variation in voters' ideological position might affect both the classification and the estimated level of

⁶ This measurement of party positions is based on Golder and Stramski (2010). As they note, this approach offers several advantages over relying on expert surveys. Most importantly, the CSES data asks citizens to place both themselves and political parties on the same left-right scale at the same point in time. This reduces problems of differential item functioning (DIF), where experts and citizens might interpret the scale differently. It also avoids validity concerns arising from differences in survey timing, scale, or phrasing.

congruence. This means that there is a possibility that part of the observed relationship is driven by the shared reliance on the scale, instead of reflecting a fully independent relationship.

Effective number of parties

To test whether the effect of negative voting on congruence differs depending on party system fragmentation, I will include the *effective number of parties (ENEP)* as an interaction term. To operationalize this, I use the CSES variable for the effective number of electoral parties that captures both the number of parties and their relative strength by weighting parties according to their vote share. This ensures that larger parties contribute more to the resulting index and that the measurement reflects the number of relevant competitors in each party system instead of counting all parties.

I argue that this operationalization better captures my theoretical expectations than a dichotomous measure of party systems (e.g., two-party vs. multiparty). The underlying argument for the interaction concerns the structure of electoral choice faced by voters, specifically the number of viable parties. By using the number of effective parties, I am, in contrast, able to distinguish between contexts with three versus eight relevant parties, which would otherwise be treated as equivalent within broader party system categories. Also, since the effective number of parties may change within countries over time, the analysis will assess both differences across countries and within countries.

To easier interpret the interaction term, the variable for ENEP and negative voting are centered around the mean value prior to estimation (Long, 2024). By doing that, the main effect of negative voting can be interpreted as its effect at the average level of party system fragmentation in the sample, rather than at a value representing zero parties, which would lack substantive meaning.

Lastly, the effective number of electoral parties is included in the baseline models to account for the direct association between party system fragmentation and congruence.

Control variables

Furthermore, to reduce the risk of omitted variable bias, I include a set of control variables that may influence both the prevalence of negative voting and the level of congruence. Given that this specific relationship has not been studied before, the selection of controls is guided by empirical findings from each respective literature. Starting with the literature on negative voting, I incorporate aggregated sociodemographic characteristics that have been shown to

systematically shape this behavior: *age* and *employment* (Garzia & Ferreira da Silva, 2022b). Age is measured as the mean age in the election and country and employment conditions are measured using the unemployment rate (% of the total labor force). In addition, I include an aggregated measure of *partisanship*, as weaker partisan ties are shown to be associated with a higher likelihood of negative voting. Partisanship is measured as party identification (yes/no) and aggregated as the share of respondents without party identification.

Moreover, religiousness and negative evaluations of the economy are included as additional robustness controls (Ibid.) However, because these variables contain substantially fewer observations, they are not included in the main analysis and are introduced separately in a later stage to check if the results remain stable. Turning to the operationalizations of these controls, religiousness is measured as the aggregated mean frequency of attendance at religious services at the country-year level. Economic evaluations are based on respondents' assessment of the national economy over the past twelve months and are operationalized as the share reporting a worsening economy.

In contrast, identifying appropriate control variables for congruence requires drawing on related literature, as direct evidence on its predictors remains limited. Most research on congruence has focused on the effects of electoral systems (e.g., Golder & Ferland, 2018; Golder & Stramski, 2010), offering less guidance on potential determinants. However, studies of unequal representation provide useful insights into relevant confounders. For example, Lupu and Warner (2022) show that domestic structural conditions, particularly economic conditions and indicators of good governance, are important for shaping representational inequality. Drawing on their results, I therefore include controls for *logged GDP per capita* and *Human Development Index (HDI)*, both measured at the election year (time t). GDP per capita is based on World Bank estimates in purchasing power parity (PPP) terms and is logged to reduce skewness, while HDI is a composite index ranging from 0 to 1 capturing average achievements in health, education and standard of living.

Important to note is that several factors highlighted in previous research as predictors of congruence, such as the quality of democracy or levels of corruption (Lupu & Warner, 2022b), have been intentionally excluded from the main analysis. The reason for this is that their position in the causal chain is theoretically unclear and could act as either exogenous confounders, endogenous mediators or moderators of the relationship. For instance, lower level democracies may have higher levels of dissatisfaction, and thus negative voting, while simultaneously producing poorer representation. However, it could also be a part of the chain

between the variables, as increased reliance on negative voting is theoretically expected to undermine accountability and weaken representative institutions over time. Controlling for such a variable would therefore risk absorbing parts of the effect the analysis aims to estimate and in turn underestimate the actual relationship. At the same time, the inclusion of more general structural indicators such as HDI should be able to capture differences in development and governance without being directly tied to the theoretical mechanism at play.

3.2 Method

Lastly, to assess the relationship between negative voting and incongruence, I employ two complementary regression models that capture different sources of variation in the data. Given that the nature of this relationship is yet to be established, the models are designed to evaluate the robustness of the result across alternative assumptions about unobserved heterogeneity.

Accordingly, the first model applies a two-way fixed-effects (FE) specification within an ordinary least squares (OLS) regression framework. This is done with the aim of strengthening causal inference by analyzing the relationship between negative voting and substantive representation from changes *within* the same political system (Angrist, 2014). By controlling for unobserved, time-invariant differences across countries, the model automatically accounts for structural differences that could otherwise confound the relationship. For instance, systematic differences in how respondents in different countries interpret the left-right scale used to measure incongruence, or institutional differences such as variation in electoral rules (e.g., majoritarian vs. proportional systems). At the same time, year effects account for global events such as an economic crisis, which most likely affects both voter behavior and patterns of representation across countries.

However, while the FE model offers important advantages in controlling for unobserved heterogeneity across countries and over time, it also involves certain limitations. Because the estimation relies exclusively on within-country variation over time, it requires sufficient temporal variation in both the independent and dependent variables. For example, it overlooks relationships that exist primarily across countries rather than within them.

In contrast, the second model is designed to allow both within- and between-country variation to contribute to the estimation of the regression coefficient (β_1). Specifically, I estimate a multilevel random intercept and slope (RE) model with year fixed effects. This is done by first introducing a country-specific intercept term, which allows each country to have its own baseline level of incongruence. In other words, each country's estimated regression line is

allowed to shift up and down depending on structural differences in the dependent variable. However, instead of removing these cross-national differences (as in the FE specification), the RE specification incorporates them into the estimation of an overall average relationship across all countries in the sample (Centre for Multilevel Modelling, n.d.). In practical terms, the RE specification treats these country-specific intercepts as randomly distributed around a common mean and all countries are modeled as part of a broader population of parliamentary democracies.

In addition, I include random slopes for the main independent variable, which also allows the estimated effect of negative voting on incongruence to vary across countries. This choice is theoretically motivated by the expectation that the relationship between negative voting and substantive representation is conditional. Together, the combined model allows countries deviate from the overall average through both its intercept and slope (Heisig & Schaeffer, 2019; Centre for Multilevel Modelling, n.d.).

At the same time, even though the model allows each country to differ in their baseline levels of incongruence and estimated effect, it still relies on the assumption that these unobserved country-specific characteristics are not systematically correlated with the explanatory variables (Centre for Multilevel Modelling, n.d.). This is a stronger assumption than in the FE model and implies that the estimated coefficients may be biased if countries with certain characteristics also systematically differ in levels of negative voting or other variables included in the model. While the inclusion of more structural control variables such as party system fragmentation and human development could partly reduce this concern, the results should be interpreted with caution.

Furthermore, estimating panel regression models requires several assumptions regarding the behavior of the error term (ϵ). One important concern is autocorrelation, which occurs when the residuals are correlated across observations rather than independent (Angrist, 2014). Since the analysis is conducted at a country-year level, the CSES data consists of repeated observations nested within countries over time where observations from the same country are likely to be correlated across elections. Failing to recognize this grouping could lead to underestimated standard errors and an overestimation of statistical significance. In the RE specification, this clustered nature is addressed through the multilevel structure of the model. The random intercept allows observations within countries to share a common baseline, while the random slopes account for unexplained cross-country variation in the main relationship (Heisig & Schaeffer, 2019; Centre for Multilevel Modelling, n.d.).

A related issue is heteroskedasticity, which may occur if the variance of the residuals either increases with the predicted value of the dependent variable or varies systematically with the independent variable (Angrist, 2014). To assess this, Breusch-Pagan tests were conducted for both model specifications (see Appendix). This test indicates heteroskedasticity across the FE models, while no clear evidence was found for the RE models. Therefore, to account for both heteroskedasticity and serial correlation, clustered standard errors at the country level are applied in the FE specification.

To address differences in measurement across observations, all models are weighted by the number of respondents in each country-year sample. Because both the dependent and independent variables are estimated using survey data, each observation represents an aggregate based on varying numbers of individual responses. Weighing, therefore, assigns greater influence to observations with more statistically precise estimates, which is particularly important given the relatively limited number of macro-level observations.

4 Results

This section presents the empirical results of this thesis in five steps. First, I provide a descriptive overview of the main variables, their empirical distribution and the bivariate relationship between them. Second, I examine the relationship between negative voting and incongruence (H1) using two-way fixed effects models, which focus on within-country variation over time. Third, I estimate random intercept and slope models to assess whether the relationship emerges when both within-and between country variation are considered. Fourth, as the more exploratory part of the analysis I include interaction terms between negative voting and the effective number of electoral parties to evaluate whether the relationship is conditioned by party system fragmentation. Finally, the section concludes with a set of robustness checks to assess the sensitivity of the results to alternative specifications and measurement choices.

4.1 Descriptive Statistics

Table 1 presents summary statistics for the dependent variable, the different operationalizations of negative voting and the control variables. Starting with the dependent variable, the average level of incongruence (EMD) is relatively low. With a mean value of 1.15 and a standard deviation of 0.39 the country-year observations generally reflect a close alignment between the distributions of citizens and representatives. As shown in Figure 1, the distribution of incongruence is moderately right skewed, with most observations concentrated between

approximately 0.8 and 1.5, with a smaller number of cases extending toward higher values above 2. This means that the empirical distribution is concentrated at the lower end, despite the scale ranging from 0-10. Even the highest observed values correspond to moderate levels of incongruence. In other words, although these higher values could be considered outliers in a strict statistical sense, they are not extreme relative to the scale and are therefore retained.

As previously discussed, the different operationalizations of negative voting vary in how many voters they classify as negative consistent with the conceptual trade-off between false positives and false negatives. The relative measure yields the lowest average (0.05), reflecting its restrictive definition, whereas the threshold and ideological deviation measures respectively capture a larger proportion of the respondents (0.27 and 0.23). Figure 1 further illustrates that the threshold and ideological deviation measures display broadly similar distributions, with observations centered around average values and variation across most of their empirical range, although also with some right skewness. This indicates that negative voting, as captured by these operationalizations, is relatively common and varies in intensity across country-years. There is no clear indication that the estimates are driven solely by a small number of extreme observations.

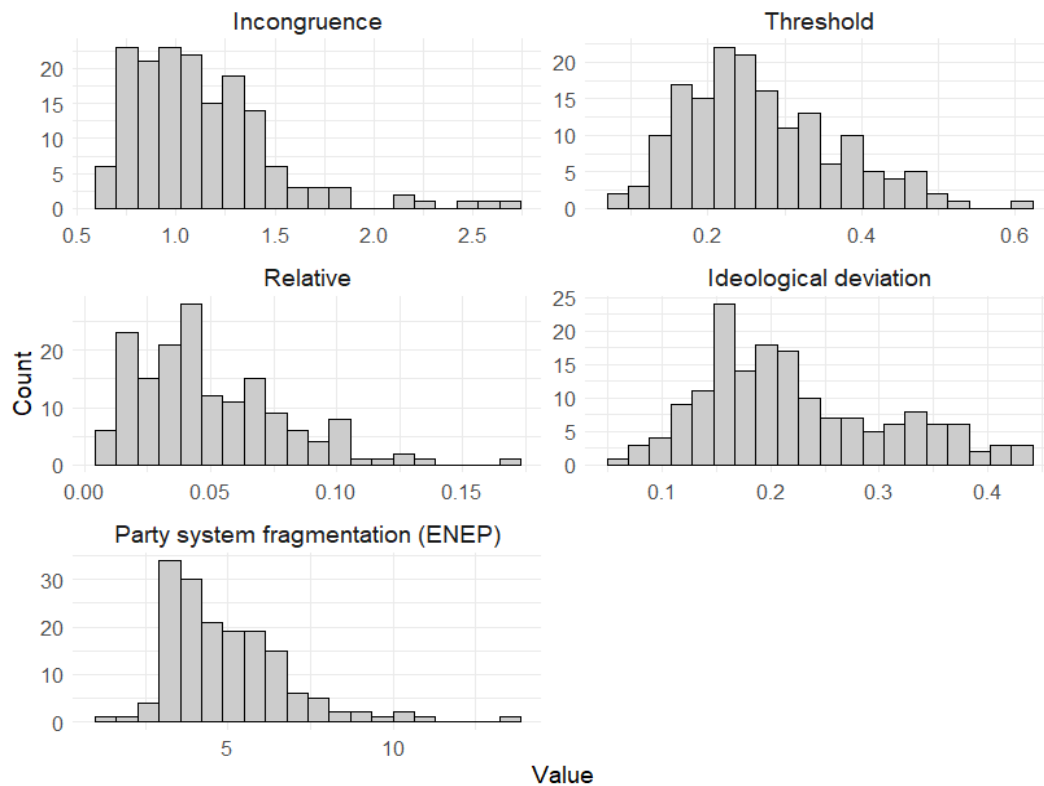
By contrast, the relative measure is heavily concentrated at low values, with most observations clustering around zero and only a small number of cases extending to higher values. As a result, variation in this measure is primarily driven by a small number of cases with higher values. Taken together, this pattern tells us that the prevalence of negative voting is highly sensitive to how the concept is operationalized, which further motivates the use of multiple measures throughout the analysis.

Finally, the structure of the party system varies considerably across the sample. The effective number of electoral parties has a mean of 5.14 and a standard deviation of 1.99, ranging from two-party systems to highly fragmented contexts with more than ten parties. As illustrated by Figure 1, most observations fall between around 3 and 7 effective parties, although a smaller number of cases extend toward substantially higher numbers. This variation provides a useful basis for examining how party system fragmentation conditions the relationship between negative voting and incongruence.

Table 1. Summary statistics of the main variables and control variables

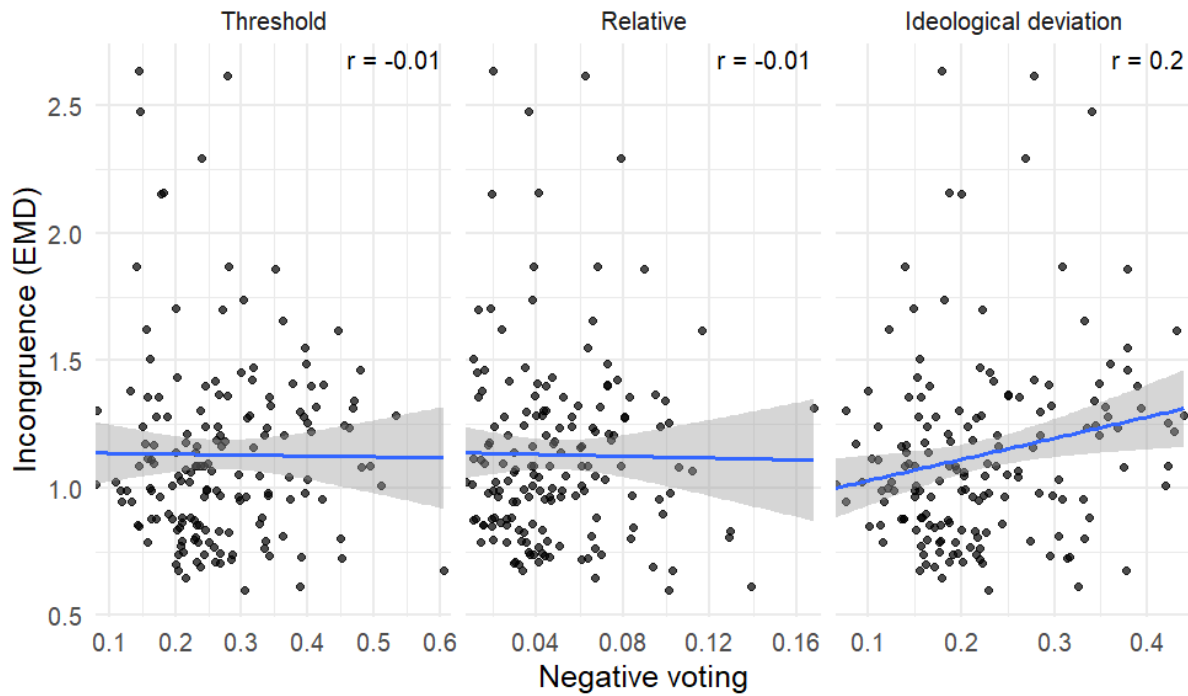
(In)dependent Variables	N	Mean	SD	Min	Max
Incongruence (EMD)	175	1.15	0.39	0	10
Negative voting: <i>Threshold</i>	179	0.27	0.10	0	1
Negative voting: <i>Relative</i>	179	0.05	0.03	0	1
Negative voting: <i>Ideological deviation</i>	179	0.23	0.09	0	1
Effective number of electoral parties	173	5.14	1.99	0	15
Control Variables					
Mean age	179	49.66	4.77	15	115
Share weak partisanship	178	0.47	0.14	0	1
Log GDP per capita	175	10.35	0.58	0	13.71
Unemployment rate (% of total labor force)	177	7.37	4.98	0	100
HDI	173	0.87	0.07	0	1
Share negative economy evaluations	110	0.36	0.23	0	1
Religiousness	130	2.72	0.81	1	6

Notes: The unit of analysis is the country-year, and the sample is restricted to observations used in the main analysis.

Figure 1. Distribution of main variables in the sample

Note: The figure presents histograms of ideological incongruence (EMD), the three operationalizations of negative voting and ENEP across the country-year observations in the main analysis. The x-axis represents the values of the variable and the y-axis the number of observations.

Figure 2. Scatterplot of the relationship between negative voting and incongruence



Note: The figure presents scatterplots illustrating the relationship between negative voting and incongruence (EMD) across the country-year observations in the main analysis. Each panel corresponds to one measure of negative voting. A linear regression line with 95% confidence intervals is included in each panel along with Pearson's correlation measure.

As a preliminary step to the main analysis, Figure 2 examines the bivariate relationship between negative voting and incongruence across the three operationalizations. This provides an initial assessment of the hypothesis that higher levels of negative voting are associated with higher levels of incongruence.

Overall, this descriptive evidence suggests that the relationship is weak and sensitive to how negative voting is measured. More specifically, in the threshold and relative panel, the fitted line is close to horizontal and the correlation measure is close to zero, indicating little systematic correlation between negative voting and incongruence. The confidence intervals are also relatively wide, especially at the extremes of the scale, further suggesting a high degree of uncertainty around the estimated slopes. In comparison, the ideological deviation measure illustrates a weakly positive relationship where observations with higher levels of negative voting also tend to have a higher value of incongruence. However, this pattern should not be

overstated either as the dispersion of observations around the fitted line remains substantial and similar levels of negative voting correspond to a wide range of incongruence values.

Certain features of the data could help explain why the observed support for the hypothesis is relatively weak. For instance, a small number of cases combining relatively low levels of negative voting with comparatively high incongruence seem to weaken the estimated association, especially for the threshold and relative measures. In simple terms, observations like these tend to pull the fitted lines downward at the lower end of the independent variable.⁷ In addition, the figure treats country-year observations as independent and does not account for the panel structure of the aggregated data. As a result, it cannot distinguish between relationships driven by differences between countries and those driven by changes within countries over time.

In summary, this bivariate analysis reveals mixed support for the effect of negative voting and incongruence. However, this analysis does not yet take into account the nested structure of the data or potential endogeneity. For that reason, I now turn to multivariate models.

4.2 Two-way Fixed Effects Models (FE)

Turning to the main results, Table 2 examines the relationship between negative voting and incongruence using two-way fixed effects models, thereby testing whether changes in negative voting within countries over time are associated with changes in substantive representation. Although all coefficients are positive, only the estimated effect for the ideological deviation measure is statistically significant ($p < 0.05$). Therefore, although increases of negative voting are associated with higher levels of incongruence in one model, the results provide only limited and inconsistent evidence in support of the hypothesis.

In terms of magnitude, the statistically significant effect is moderate. In order to facilitate substantive interpretation and comparison across operationalizations, the estimated coefficients are interpreted relative to a one standard deviation increase. Followingly, given that the measure has a standard deviation of approximately 0.09, a one standard deviation increase in negative voting is associated with an increase of roughly 0.11 units on the EMD scale ($1.209 \times 0.09 =$

⁷ At the same time, these observations are substantively meaningful and likely reflect real variation in how electoral behavior translates into representation across different contexts. Excluding them would therefore risk introducing bias by selectively removing cases that do not fit the general pattern. For this reason, they are kept in the analysis.

0.109). In other words, this effect translates to about one third of the change in the dependent variable.

This inconsistent pattern is further illustrated in the predicted values plots in Figure 3. While all models display a positive relationship between negative voting and incongruence, the predicted lines for the threshold and relative measures are accompanied by statistical uncertainty. In these two panels the confidence intervals are wide enough to include a null effect ($\beta_1 = 0$), indicating that the true relationship may be indistinguishable from zero. This illustrates why the estimated coefficients fail to reach statistical significance.

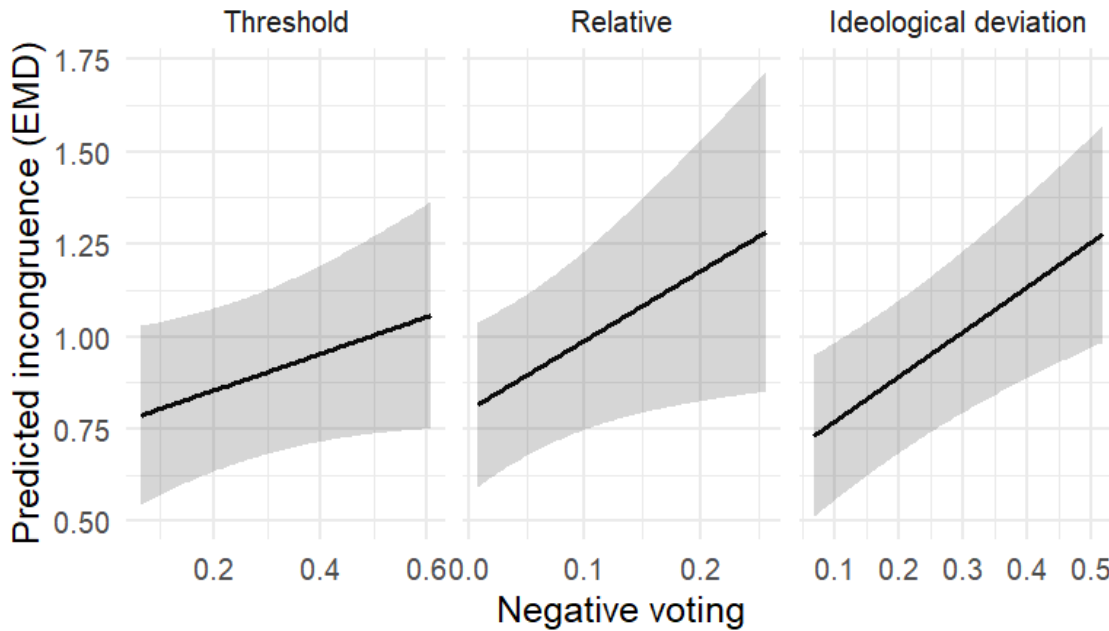
At the same time, the inconsistent effects should be interpreted in light of the model specification. The adjusted R^2 values are very high (0.84-0.86) which indicates that the country and year fixed effects account for a large share of the variation in incongruence. This interpretation is further supported by the fact that most of the control variables fail to reach statistical significance, despite being theoretically relevant predictors of congruence (Garzia & Ferreira da Silva, 2022b; Lupu & Warner, 2022b). By conducting an additional comparison, the high explanatory power of the FE models is primarily driven by the inclusion of country-fixed effects. Models including only year fixed effects explain less variation (adj R^2 between 0.65-0.67) than models including country fixed effects (adj R^2 between 0.83-0.84), although both values are relatively high. Consequently, when most of the variation in the data comes from differences between countries, the fixed-effects estimator has little remaining variation to explain.

Table 2. Two-way fixed effects models

Model	Incongruence (EMD)		
	Threshold	Relative	Ideological deviation
Negative voting	0.499 (0.529)	1.878 (1.452)	1.209* (0.548)
Mean age	0.018 (0.016)	0.015 (0.013)	0.019 (0.013)
Share weak partisanship	-0.120 (0.217)	-0.109 (0.209)	-0.131 (0.200)
Log GDP per capita	0.308. (0.179)	0.258. (0.152)	0.335* (0.162)
Unemployment	0.007 (0.011)	0.013 (0.011)	0.004 (0.010)
HDI	0.561 (1.978)	0.632 (1.831)	0.846 (1.870)
Effective number of electoral parties	-0.032 (0.023)	-0.037 (0.023)	-0.035 (0.022)
Time FE	✓	✓	✓
Country FE	✓	✓	✓
N	164	164	164
Adj R ²	0.84	0.85	0.86

Notes: Clustered standard errors in parentheses. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$. The models include country and year fixed effects, clustered standard errors at the country level and are weighted by country-year sample size.

Figure 3. Predicted values of incongruence by levels of negative voting in the FE models



Note: The figure illustrates predicted values from the FE models across the three operationalizations of negative voting. Each panel shows the estimated relationship between negative voting and incongruence (EMD) while holding all control variables constant. A linear regression line with 95% confidence intervals is included in each panel.

4.3 Random Intercept and Slope Models (RE)

Table 3 presents the results from the random intercept and slope model (RE), examining the average estimated relationship between negative voting and incongruence when both between- and within-country variations are considered. In contrast to the results presented above, the estimated coefficients in Table 3 for negative voting are consistently positive and statistically significant for the relative ($p < 0.01$) and ideological deviation measures ($p < 0.001$). Accordingly, the comparison between fixed and random effects specifications provides mixed evidence in favor of the main hypothesis.

More specifically, based on this comparison, I highlight two main findings. First, the relationship between negative voting and incongruence is not equally robust across model specifications and depends on how variation in the data is handled. The absence of an effect in most of the FE models, combined with the stronger and more consistent effects observed in the RE model, suggests that the relationship is driven mainly by cross-national comparison rather than changes within countries over time. When holding other included variables constant, this

means that countries with higher levels of negative voting also tend to have higher levels of incongruence (EMD).

Second, turning to the estimated regression coefficients more directly, the different results across operationalizations indicate that the way negative voting is measured plays an important role in shaping empirical results. Based on the results in Table 2 and 3, we can reject the null hypothesis of no relationship ($\beta_1=0$) between negative voting and incongruence for the ideological deviation measures in both model specifications and the relative measure in the RE model. However, we fail to reject it for the threshold measure. This suggests that the relationship of interest is most evident when negative voting is measured either as a relative evaluative profile or as a combination of negative affect and deviation from proximity voting.

Although the estimated relationship is more consistent in the RE models, the magnitude of the effect is somewhat weaker than in the FE specification. In Table 3, for the relative measure, an increase of one standard deviation in negative voting is associated with an average increase of around 0.07 units on the EMD scale ($2.203 \times 0.03 = 0.066$). Similarly, for the ideological deviation measure, with a standard deviation of 0.09, a one standard deviation increase corresponds to an average increase of roughly 0.08 units ($0.932 \times 0.09 = 0.084$). For both measures, this corresponds to about one fifth of the change in the dependent variable.

Lastly, turning to the control variables, the results largely align with expectations from previous literature. Both the Human Development Index and the effective number of parties are statistically significant and negatively associated with incongruence across all models. Substantively, this means that higher levels of socio-economic development and more fragmented party systems are associated with lower levels of incongruence (i.e., better congruence). These findings are consistent with prior research suggesting that more developed democracies (Lupu & Warner, 2022b) and systems with a larger number of parties provide better conditions for substantive representation (Golder & Ferland, 2018; Golder & Stramski, 2010; Soroka & Wlezien, 2015). In addition, the mean age of the electorate is also negative and statistically significant across all three models, indicating that country-years with older electorates tend to exhibit slightly lower levels of incongruence.

However, one control variable yields a less intuitive result. The estimated effect of the share of weak partisans is negative and significant in all three models, indicating that country-years with a higher proportion of voters without strong party identification tend to exhibit lower incongruence. This does not necessarily align with the expectation that weaker party ties may be associated with higher levels of negative voting (see Garzia & Ferreira da Silva, 2022b) and

weaker representation. One possible explanation for this could be that weaker partisanship increases electoral flexibility, allowing voters to select parties closer to their preferences.

Table 3. Random intercept and slope models

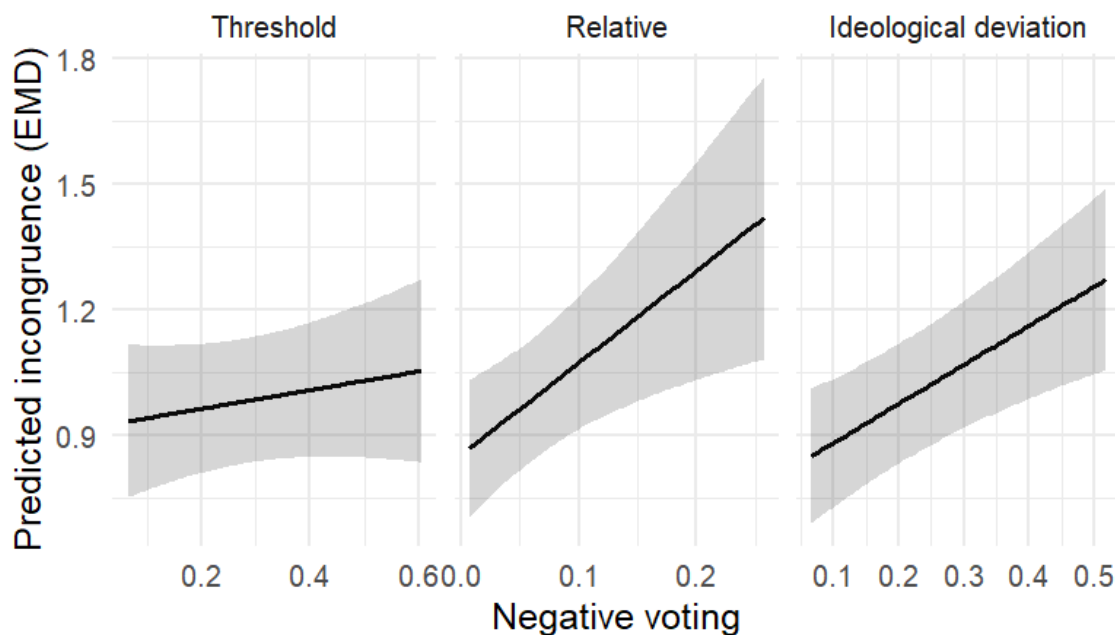
Model	Incongruence (EMD)		
	Threshold	Relative	Ideological deviation
Negative voting	0.222 (0.249)	2.203** (0.781)	0.932*** (0.266)
Mean age	-0.014* (0.006)	-0.017** (0.006)	-0.013* (0.006)
Share weak partisanship	-0.486** (0.155)	-0.530*** (0.147)	-0.577*** (0.147)
Log GDP per capita	-0.140 (0.113)	-0.161 (0.110)	-0.125 (0.108)
Unemployment	-0.003 (0.006)	0.000 (0.005)	-0.005 (0.005)
HDI	-2.176* (0.855)	-1.970* (0.823)	-2.200** (0.810)
Effective number of electoral parties	-0.049*** (0.014)	-0.052*** (0.014)	-0.048*** (0.013)
Time FE	✓	✓	✓
Random Intercept	✓	✓	✓
Random Slope	✓	✓	✓
N	164	164	164

Notes: Standard errors in parentheses. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$. Models include year fixed-effects, a random intercept for countries, a random slope for negative voting and are weighted by country-year sample size.

Figure 4 graphically illustrates the average estimated relationship between negative voting and incongruence based on the RE models in Table 3. Across all three panels, the predicted relationship is positive, indicating that elections with higher shares of negative voting also tend to have higher levels of incongruence. However, the strength and precision of the relationship differ across the three measures. The relative and ideological deviation panels show steeper slopes compared to the threshold one, indicating a stronger substantive effect on incongruence.

The width of the confidence intervals further provides important information about statistical uncertainty. The shaded areas around the predicted lines represent 95 percent confidence intervals. For the threshold measure, this area is relatively wide and overlaps considerably across different values of negative voting, which explains why the corresponding coefficient is not statistically significant. By contrast, the relative and ideological deviation measures show comparatively clearer positive relationships, although the precision differs between the two operationalizations. For the relative measure, the confidence intervals widen considerably at higher levels of negative voting, reflecting greater uncertainty at the upper end of the distribution where fewer observations are available. The ideological deviation measure, in comparison, displays a more stable level of uncertainty across its observed range.

Figure 4. Predicted values of incongruence by levels of negative voting in the RE models



Notes: The figure illustrates predicted values from the RE models across the three operationalizations of negative voting. Each panel shows the average estimated relationship between negative voting and incongruence (EMD) when holding all control variables constant. A linear regression line with 95% confidence intervals is included in each panel.

4.4 Test of Interaction Effects

So far, the analysis has allowed the effect of negative voting to vary across countries by estimating random slope models. However, this variation has yet to be explicitly linked to the more exploratory question of this thesis, where the effect is expected to be conditioned by party

system fragmentation. Due to inconsistent results in the FE specification and its reliance on limited within-country variation in the dependent variable, the interaction term between negative voting and ENEP is only included in the RE specification.

In terms of interpretation, table 4. presents the estimated coefficient of each negative voting measure represents the effect of negative voting when ENEP is at its sample mean. Conversely, the estimated coefficient for ENEP illustrates its effect when negative voting is at its average. The interaction term captures how the marginal effect of negative voting changes when ENEP increases.

Turning to the analysis, the results provide no support for a moderating effect of party system fragmentation. Across all three models in Table 4, the interaction terms between negative voting and ENEP are positive but not statistically significant ($p > 0.1$). This means that we cannot reject the null hypothesis that the marginal effect of negative voting on incongruence is constant across different numbers of effective parties in the population that we aim to estimate.

Looking beyond the regression coefficients, Figure 5 provides a more detailed picture of the estimated relationships. The two lines represent the predicted average effect of negative voting across two different levels of ENEP, low and high, measured as one standard deviation below the sample mean and one standard deviation above the sample mean. In all three models, the relationship between negative voting and incongruence remains positive across different ENEP levels, particularly for the relative and the ideological deviation measures. However, the slopes of the predicted lines remain largely parallel, indicating that the marginal effect of negative voting is relatively similar across different party systems.

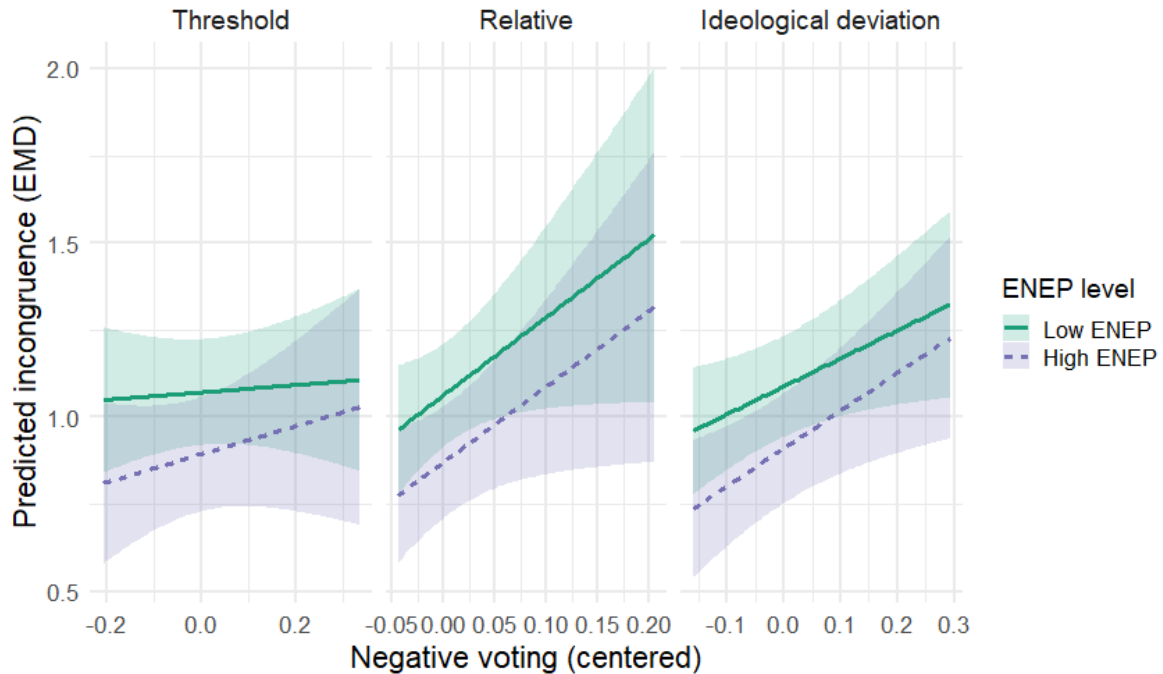
These findings are further reinforced by statistical uncertainty. The confidence intervals surrounding the predicted lines overlap substantially across ENEP levels, especially at higher levels of negative voting where observations are more limited. This overlap indicates that the differences in slopes are not statistically distinguishable from one another, which is consistent with the non-significant interaction terms reported in Table 4.

Table 4. Random intercept and slope interaction models

Incongruence (EMD)			
Model	Threshold	Relative	Ideological deviation
Negative voting (centered)	0.255 (0.257)	2.206** (0.787)	0.943*** (0.268)
ENEP (centered)	-0.048** (0.014)	-0.052*** (0.014)	-0.048*** (0.013)
Negative voting $\times$ ENEP	0.080 (0.155)	-0.020 (0.426)	0.076 (0.155)
Mean age	-0.013* (0.007)	-0.017* (0.006)	-0.012. (0.006)
Share weak partisanship	-0.477** (0.156)	-0.531*** (0.150)	-0.560*** (0.151)
Log GDP per capita	-0.146 (0.114)	-0.160 (0.111)	-0.129 (0.109)
Unemployment	-0.003 (0.006)	0.000 (0.005)	-0.004 (0.005)
HDI	-2.144* (0.859)	-1.973* (0.829)	-2.164** (0.816)
Time FE	✓	✓	✓
Random Intercept	✓	✓	✓
Random Slope	✓	✓	✓
N	164	164	164

Notes: Standard errors in parentheses. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$. Models include year dummies, a random intercept for countries, and a random slope for negative voting and are weighted by country-year sample size. All interaction terms are estimated using mean-centered variables.

Figure 5. Predicted incongruence by levels of negative voting across party system fragmentation



Note: The figure presents predicted values from the random intercept and slope interaction models across the three operationalizations of negative voting. Each panel shows the average estimated relationship between negative voting (centered) and incongruence (EMD) at different levels of party system fragmentation (ENEP), when holding all control variables constant. Low ENEP is measured as one standard deviation below the sample mean and high ENEP to one standard deviation above the sample mean. Around each linear regression line is a 95% confidence interval.

4.5 Robustness Checks and Extended Analyses

To validate the results of this thesis, three robustness checks are conducted (a more detailed presentation of these is found in the Appendix). These tests assess the sensitivity of the findings to alternative measurements and additional control variables.

First, to ensure that the findings are not driven by the specific operationalization of congruence, I compare the Earth Mover's distance (EMD) and the cumulative distribution function used by Golder and Stramski (2010). This association is evaluated using a Pearson correlation coefficient at the country-year level. The two measures are almost perfectly correlated ($r = 0.98$), indicating that they capture nearly identical patterns of the many-to-many congruence across contexts. While this is expected given that both measures rely on distributional distances using the left-right scale, it nevertheless provides important reassurance that the results are not a consequence of a specific methodological choice.

Second, following Garzia and Ferreira da Silva (2022b) I include both religiousness and negative evaluations of the economy as additional control variables to assess whether the main results are sensitive to their omission. The inclusion of religiousness leaves the results largely unchanged across both model specifications and operationalization. By contrast, including negative economic evaluations leads to more notable changes. The estimated effect for the ideological deviation measure loses statistical significance in the FE specification, while both the relative and the ideological deviation measure remain statistically significant in the RE specification, although at a lower level ($p < 0.05$).

These changes could either reflect the inclusion of the additional control variable itself or the substantial reduction of sample size. The estimated coefficient for negative economic evaluations fails to reach conventional levels of statistical significance in any of the models. However, while these findings suggest that the control does not independently explain variation in incongruence, its inclusion could still affect the estimates indirectly if it is correlated with other variables. Additionally, the inclusion of negative economic evaluations substantially reduces the sample size from 164 to 99 observations due to missing values. Fewer observations reduce the precision of the estimation and increase uncertainty around the coefficients. Consequently, the weaker results may partly reflect lower statistical power or systematic differences between the reduced and original samples.

To test this, I re-estimated the models based on the same restricted sample without the inclusion of the additional control. When doing so, the results remain largely unchanged compared to the models including the controls, indicating that the weakening of the results is driven primarily by the reduced sample size.

Lastly, because interactions with two continuous variables require substantial statistical power, additional tests were conducted using ENEP as both a two- and three-category variable. Interestingly, these alternative specifications produce different results from the main analysis and provide some support for a conditional effect. In both tests, the interaction terms are consistently negative suggesting that the relationship between negative voting and incongruence is substantially weaker in multiparty systems than in two-party systems. These are also statistically significant for two of the operationalizations of negative voting ($p < 0.05$ when ENEP is binary and $p < 0.01$ when ENEP is measured in three categories). The three-category specification further illustrates that the weaker relationship emerges already in moderately fragmented party systems and that the differences between moderate and highly fragmented systems are comparatively small. This suggests that the conditional effect relates

more to the distinction between two-party and multiparty systems than to increasing levels of fragmentation itself.

However, these results are inconsistent across operationalizations of negative voting and should therefore be interpreted with caution. Notably, the threshold measure becomes significant once the interaction is included. This tells us that its average effect in the baseline models may conceal substantial variation across party systems. Its positive relationship with incongruence appears concentrated in two-party systems and weakens in multiparty contexts.

Overall, these results reinforce the main patterns observed in the analysis and indicate that the empirical support for the main hypothesis is conditional rather than general. Instead of finding a uniform relationship across all specifications, the evidence indicates that the link between negative voting and incongruence is highly dependent on how the concept is measured and how the variation in the data is modeled.

5 Conclusions

As the first part of the analysis, this thesis hypothesized that higher levels of negative voting are associated with lower levels of substantive representation. As a second and more exploratory part of the analysis, the thesis also asked whether this relationship is conditional on the number of effective parties in the party system. To answer this, the analysis used post-election survey data from the CSES, covering 164 elections in 44 parliamentary countries between 1996 and 2024 and compared three operationalizations of negative voting across two model specifications.

Turning to the main hypothesis, the findings provide mixed evidence that increased negative voting is associated with weaker substantive representation, measured as incongruence between the ideological distribution of citizens and their representatives. The two-way fixed effects models provide limited and inconsistent evidence that changes in negative voting *within* countries *over time* are systematically associated with changes in incongruence. By contrast, the random intercept and slope models, that also incorporate cross-national differences into the estimation, reveal a positive and statistically significant relationship for two out of three operationalizations of negative voting. Consequently, this divergence suggests that the identified effect is neither robust across model specifications nor measurements on negative voting.

Accordingly, rather than pointing to a single clear conclusion, I choose to highlight two broader and theoretically meaningful patterns. First, the divergence between the two models specifications suggests that the relationship, if present, is driven primarily by cross-national differences rather than short-term changes within countries. This is an important finding in relation to the broader theoretical starting point of this thesis. If negative voting is mainly linked to differences between countries, it may be less useful to understand it mainly as a short-term consequence of rising affective polarization. Instead, the results indicate that negative voting may be embedded in more stable structural or cultural characteristics.

The second important finding is that the estimated relationship depends strongly on how negative voting is conceptualized and measured. The effect is most consistently captured when negative voting is operationalized as a combination of negative affect, neutrality toward the party choice and deviations from proximity voting. This operationalization is statistically significant across both specifications, whereas the relative measure emerges first when between-country variation contributes to the estimation. This further suggests that the representational consequence is most persistent when dislike is connected to electoral choices that no longer reflect voters' own ideological preferences. In this sense, the ideological deviation measure might come closer to capturing actual negative voting behavior, because it identifies cases where opposition is not only present as an attitude but also reflected in a departure from proximity-based voting. At the same time, the fact that the results differ so clearly across operationalizations, even when they build on similar conceptual foundations, points to a broader limitation in the existing literature. Since negative voting lacks a standardized measurement, empirical conclusions remain difficult to compare across studies.

Moving on to the second more exploratory part of the analysis, this thesis provides no evidence that party system fragmentation conditions the relationship between negative voting and substantive representation. The interaction terms fail to reach statistical significance and the predicted slopes remain parallel across different ENEP levels. Instead, additional robustness checks provide some support that the relationship differs more broadly between two-party and multiparty systems. More specifically, the positive relationship between negative voting and incongruence appears stronger in two-party systems and weaker across multiparty contexts. In comparison, the differences between moderately and high fragmented systems are comparatively small. These findings suggest that the relationship may depend less on party system fragmentation per se and more on broader institutional differences between electoral systems, which warrants further investigation.

Importantly, these findings should be interpreted in the light of several limitations. First and foremost, the observational nature of the data means that the analysis cannot establish a definite causal effect. The random effects models allow both within and between country variation to contribute to the estimation, but they also rely on the assumption that unobserved country-level characteristics are not systematically correlated with negative voting. This assumption cannot be fully guaranteed, and the results may therefore reflect systematic differences between countries where negative voting is more common. In other words, the analysis does not provide a clean counterfactual scenario of what representation would have looked like in the absence of negative voting. Additionally, reverse causality is also possible, as lower levels of representation themselves could contribute to political dissatisfaction and voting negatively. Consequently, the result should be viewed merely as first evidence of a potential relationship.

The operationalizations of negative voting also involve important trade-offs that affect how the results should be understood. The indirect approach allows for broad cross-national comparison and external validity, but at the cost of internal validity. They likely include both false negatives and false positives that might over- or underestimate the effect. Also, the shared ideological dimension between the independent and dependent variables introduces a risk that part of the observed relationship reflects mechanical correlation rather than a fully substantive effect. This reinforces the need to interpret the results as suggestive rather than conclusive.

Lastly, the relatively small number of observations limits the statistical power in the analysis, particularly in the interaction models. Consequently, the absence of a statistically significant interaction effect should not necessarily be interpreted as evidence that party system fragmentation is irrelevant for the relationship between negative voting and substantive representation.

Despite these and other limitations, the findings presented in this thesis point to several important implications that warrant further investigation. Most fundamentally, they suggest that negative voting can alter the relationship between citizens and their electoral choices. Taking it one step further, if negative voting undermines what Stimson et al. (1994) describe as electoral connection and rational anticipation between voters and political elites, such developments may also create opportunities for strategic political behavior. When electoral success depends less on policy anticipation and more on mobilizing hostility toward political opponents, parties might benefit more from symbolic opposition, polarization and identity-based hostility rather than coherent policy programs. As implicated by previous research, this might transform the political arena in the long run, in favor of populist and anti-establishment that relies more

heavily on conflict (Pereira, 2019; Weber, 2021). Also, without a clear mandate for the party or candidate, there is an increased risk of moral hazard. The incumbent have the possibility to take any action and might even assume that it is the voters preference to do so (Sulmont & Incio, 2025).

Turning to future research, there are several avenues that could extend the findings of this thesis. First, future studies should move beyond the left-right scale as a measurement of congruence. While widely used, reducing substantive representation to a single ideological dimension risks overlooking important disagreement across issues such as immigration, cultural values, or climate policy (Lupu et al., 2017). For instance, it could be the case that negative voting affects different dimensions unequally, particularly in polarized contexts where affective conflict is concentrated around specific issues rather than general ideology.

Second, more in-depth studies of single countries, outside of the United States, could help provide a clearer understanding of *how* negative voting operates within specific institutional and political contexts. Such analyses would also make it possible to apply more direct measures of negative voting and explore the mechanisms linking negative voting to representation in greater detail by, for example, also looking at responsiveness through actual policy implementations.

Finally, the findings also highlight the need for future research on the measurement of negative voting itself. As demonstrated throughout the analysis, one of the clearest findings of the analysis is that the empirical conclusions vary depending on how the concept is measured, despite a similar conceptual foundation. This raises broader concerns regarding comparability across existing studies and highlights the need for a standardized measurement.

6 References

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7 Declaration of AI

In this thesis, AI-based tools (ChatGPT and Grammarly) have been used to improve grammar and language. ChatGPT was also used as a discussion partner during the development of the research design, as help to generate and troubleshoot code in R and to understand statistical procedures and outputs. However, no part of the AI-generated output was directly incorporated into the thesis (with an exception for small grammatical corrections by Grammarly) and I take full responsibility for the content of this thesis, including all final methodological decisions, data analysis and interpretations.

Appendix A – Main Analysis

Table A.1 Breusch-Pagan tests

Model type	Model	BP statistic	p-value	Interpretation
Two-way FE	Threshold	192695.004	0.000	Evidence of heteroskedasticity
Two-way FE	Relative	192708.079	0.000	Evidence of heteroskedasticity
Two-way FE	Ideological threshold	192703.744	0.000	Evidence of heteroskedasticity
Two-way FE interaction	Threshold	192697.789	0.000	Evidence of heteroskedasticity
Two-way FE interaction	Relative	192707.833	0.000	Evidence of heteroskedasticity
Two-way FE interaction	Ideological threshold	192712.421	0.000	Evidence of heteroskedasticity
RE random intercept/slope	Threshold	3.618	0.057	No clear evidence of heteroskedasticity
RE random intercept/slope	Relative	2.676	0.102	No clear evidence of heteroskedasticity
RE random intercept/slope	Ideological threshold	2.284	0.131	No clear evidence of heteroskedasticity
RE interaction	Threshold	4.069	0.044	Evidence of heteroskedasticity
RE interaction	Relative	2.653	0.103	No clear evidence of heteroskedasticity
RE interaction	Ideological threshold	2.428	0.119	No clear evidence of heteroskedasticity

Notes: The Breusch-Pagan test evaluates the null hypothesis of homoskedastic residuals. A p-value below $\alpha = 0.05$ indicates evidence of heteroskedasticity.

Table A.2. FE models with year fixed effects only

Incongruence (EMD)			
Model	Threshold	Relative	Ideological deviation
Negative voting	0.222 (0.304)	2.203** (0.803)	0.932** (0.303)
Mean age	-0.014. (0.007)	-0.017* (0.007)	-0.013. (0.007)
Share weak partisanship	-0.486* (0.242)	-0.530* (0.214)	-0.577** (0.206)
Log GDP per capita	-0.140 (0.120)	-0.161 (0.115)	-0.125 (0.104)
Unemployment	-0.003 (0.008)	0.000 (0.007)	-0.005 (0.007)
HDI	-2.176* (1.002)	-1.970* (0.937)	-2.200* (0.861)
Effective number of electoral parties	-0.049* (0.020)	-0.052** (0.018)	-0.048** (0.018)
Time FE	✓	✓	✓
N	164	164	164
Adj R ²	0.65	0.67	0.68

Notes: Clustered standard errors in parentheses. Significance levels: *** p < 0.001, ** p < 0.01, * p < 0.05, . p < 0.10. The models include country and year fixed effects, clustered standard errors at the country level and are weighted by country-year sample size.

Table A.3 FE models with country fixed effects only

Incongruence (EMD)			
Model	Threshold	Relative	Ideological deviation
Negative voting	0.372 (0.405)	0.542 (1.163)	0.887 (0.576)
Mean age	0.003 (0.013)	0.000 (0.013)	0.004 (0.013)
Share weak partisanship	-0.115 (0.179)	-0.082 (0.193)	-0.132 (0.159)
Log GDP per capita	0.160* (0.078)	0.158. (0.082)	0.159* (0.073)
Unemployment	0.003 (0.010)	0.005 (0.009)	0.001 (0.009)
HDI	-0.796 (1.831)	-0.624 (1.765)	-0.832 (1.711)
Effective number of electoral parties	-0.035 (0.021)	-0.037 (0.023)	-0.037. (0.022)
Country FE	✓	✓	✓
N	164	164	164
Adj R ²	0.83	0.83	0.84

Notes: Clustered standard errors in parentheses. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$. The models include country and year fixed effects, clustered standard errors at the country level and are weighted by country-year sample size.

Appendix B – Robustness Checks

Table A.4. FE models with religiousness as additional control

Model	Incongruence (EMD)		
	Threshold	Relative	Ideological threshold
Negative voting	0.899 (0.677)	1.762 (1.925)	1.437* (0.681)
Mean age	0.016 (0.012)	0.018 (0.014)	0.016 (0.011)
Share weak partisanship	-0.146 (0.229)	-0.068 (0.243)	-0.117 (0.220)
Log GDP per capita	0.315 (0.247)	0.313 (0.237)	0.339 (0.224)
Unemployment	0.016 (0.011)	0.021 (0.013)	0.014 (0.010)
HDI	1.315 (2.138)	0.840 (2.150)	1.665 (2.007)
Effective number of electoral parties	-0.030 (0.028)	-0.039 (0.029)	-0.030 (0.026)
Mean religiousness	0.108 (0.084)	0.081 (0.075)	0.073 (0.081)
Time FE	✓	✓	✓
Country FE	✓	✓	✓
N	120	120	120
Adj R ²	0.87	0.87	0.89

Notes: Clustered standard errors in parentheses. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$. The models include country and year fixed effects, clustered standard errors at the country level and are weighted by country-year sample size.

Table A.5. RE models with religiousness as additional control

Incongruence (EMD)			
Model	Threshold	Relative	Ideological threshold
Negative voting	0.350 (0.276)	3.233** (0.965)	1.027*** (0.289)
Mean age	-0.010 (0.007)	-0.010 (0.007)	-0.007 (0.007)
Share weak partisanship	-0.567** (0.175)	-0.595*** (0.164)	-0.648*** (0.166)
Log GDP per capita	-0.243. (0.143)	-0.243. (0.137)	-0.214 (0.137)
Unemployment	-0.005 (0.007)	0.000 (0.006)	-0.005 (0.006)
HDI	-1.665 (1.089)	-1.870. (1.035)	-1.744. (1.024)
Effective number of electoral parties	-0.047** (0.016)	-0.055*** (0.015)	-0.047** (0.015)
Mean religiousness	0.033 (0.033)	-0.001 (0.034)	0.031 (0.032)
Time FE	✓	✓	✓
Random Intercept	✓	✓	✓
Random Slope	✓	✓	✓
N	120	120	120

Notes: Standard errors in parentheses. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$. Models include year dummies, a random intercept for countries, and a random slope for negative voting and are weighted by country-year sample size.

Table A.6 FE models with negative economy evaluations as additional control

Model	Incongruence (EMD)		
	Threshold	Relative	Ideological deviation
Negative voting	0.526 (0.891)	2.461 (2.249)	1.093 (0.903)
Mean age	0.008 (0.024)	-0.001 (0.018)	0.007 (0.021)
Share weak partisanship	-0.182 (0.344)	-0.212 (0.310)	-0.198 (0.310)
Share negative economy evaluations	0.067 (0.315)	0.017 (0.294)	0.065 (0.276)
Log GDP per capita	0.392. (0.205)	0.310 (0.186)	0.429. (0.211)
Unemployment rate	0.029 (0.021)	0.035 (0.023)	0.025 (0.019)
HDI	-1.804 (2.206)	-2.071 (2.023)	-1.964 (2.138)
Effective number of electoral parties	-0.056 (0.046)	-0.057 (0.042)	-0.057 (0.042)
Time FE	✓	✓	✓
Country FE	✓	✓	✓
N	99	99	99
Adj R ²	0.87	0.87	0.89

Notes: Clustered standard errors in parentheses. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$. The models include country and year fixed effects, clustered standard errors at the country level and are weighted by country-year sample size.

Table A.7 RE models with negative economy evaluations as additional control

Model	Incongruence (EMD)		
	Threshold	Relative	Ideological deviation
Negative voting	0.255 (0.318)	2.194* (1.044)	0.861* (0.326)
Mean age	-0.008 (0.008)	-0.012 (0.008)	-0.007 (0.008)
Share weak partisanship	-0.421* (0.206)	-0.428* (0.195)	-0.496* (0.195)
Share negative economy evaluations	0.258 (0.189)	0.201 (0.186)	0.302 (0.181)
Log GDP per capita	0.078 (0.148)	0.031 (0.145)	0.107 (0.142)
Unemployment	0.000 (0.007)	0.006 (0.007)	-0.001 (0.007)
HDI	-4.710*** (1.190)	-4.271*** (1.139)	-4.690*** (1.118)
Effective number of electoral parties	-0.054** (0.017)	-0.056** (0.016)	-0.050** (0.016)
Time FE	✓	✓	✓
Random Intercept	✓	✓	✓
Random Slope	✓	✓	✓
N	99	99	99

Notes: Standard errors in parentheses. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$. Models include year dummies, a random intercept for countries, and a random slope for negative voting and are weighted by country-year sample size.

Table A.9. FE models on restricted sample without negative economic evaluations as control

Incongruence (EMD)			
Model	Threshold	Relative	Ideological deviation
Negative voting	1.013 (1.344)	2.947 (3.420)	1.720 (1.262)
Mean age	0.009 (0.022)	0.013 (0.030)	0.001 (0.020)
Share weak partisanship	-0.219 (0.352)	-0.105 (0.354)	-0.209 (0.298)
Log GDP per capita	0.255 (0.295)	0.235 (0.230)	0.317 (0.264)
Unemployment	0.029 (0.030)	0.029 (0.033)	0.031 (0.027)
HDI	-0.794 (2.305)	-0.708 (2.326)	-1.079 (2.303)
Effective number of electoral parties	-0.061 (0.055)	-0.073 (0.064)	-0.050 (0.046)
Time FE	✓	✓	✓
Country FE	✓	✓	✓
N	99	99	99
Adj R ²	0.89	0.89	0.91

Notes: Clustered standard errors in parentheses. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$. The models are estimated on the restricted sample used in the religiousness robustness check, but without including religiousness or negative economic evaluations as control variables.

Table A.8. RE models on restricted sample without negative economic evaluations as control

Model	Incongruence (EMD)		
	Threshold	Relative	Ideological deviation
Negative voting	0.262 (0.319)	2.366* (1.033)	0.813* (0.329)
Mean age	-0.008 (0.008)	-0.012 (0.008)	-0.007 (0.008)
Share weak partisanship	-0.391. (0.206)	-0.409* (0.194)	-0.454* (0.196)
Log GDP per capita	0.014 (0.141)	-0.021 (0.137)	0.031 (0.136)
Unemployment	0.003 (0.007)	0.008 (0.007)	0.003 (0.007)
HDI	-4.224*** (1.143)	-3.879** (1.081)	-4.109*** (1.075)
Effective number of electoral parties	-0.056** (0.017)	-0.057** (0.016)	-0.053** (0.016)
Time FE	✓	✓	✓
Random Intercept	✓	✓	✓
Random Slope	✓	✓	✓
N	99	99	99

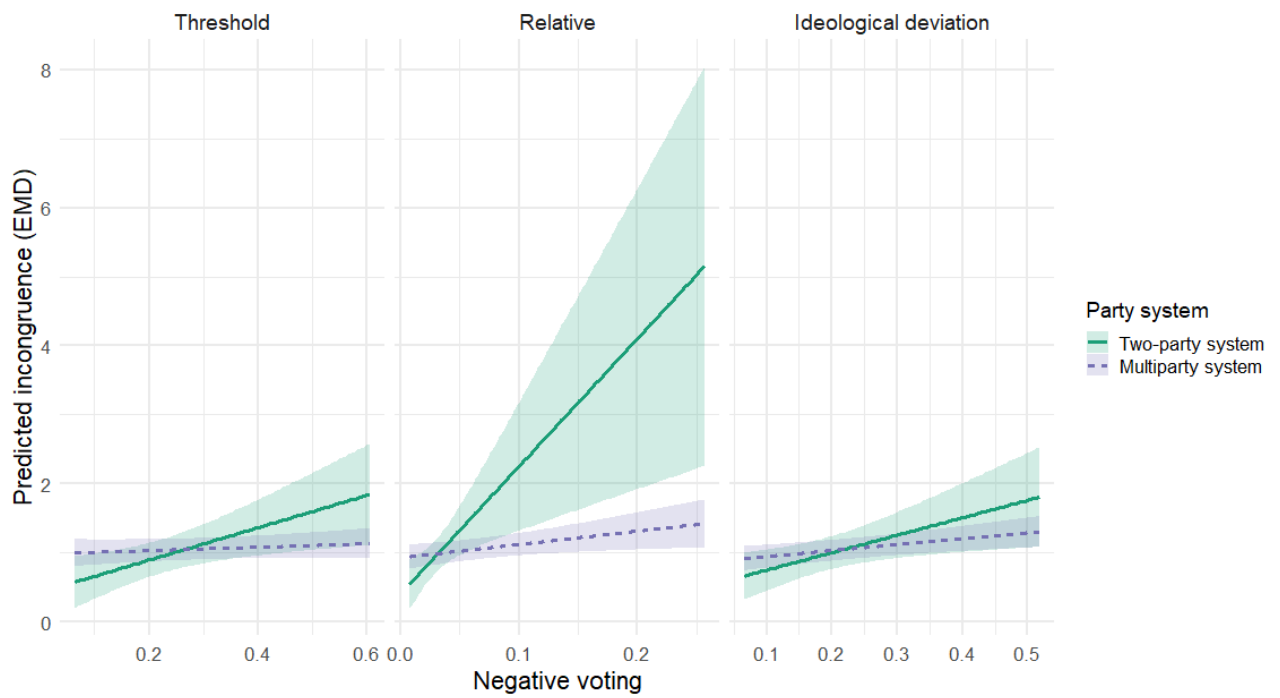
Notes: Standard errors in parentheses. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$. The models are estimated on the restricted sample used in the negative economic evaluations robustness check, but without including negative economic evaluations as a control variable.

Table A.10 RE interaction models with binary ENEP

Incongruence (EMD)			
Model	Threshold	Relative	Ideological threshold
Negative voting	2.343* (0.914)	18.482** (6.446)	2.531* (1.050)
Multiparty system	0.558* (0.240)	0.528* (0.222)	0.366 (0.242)
Negative voting × Multiparty system	-2.095* (0.926)	-16.562* (6.459)	-1.673 (1.087)
Mean age	-0.012. (0.007)	-0.016* (0.007)	-0.011. (0.006)
Share weak partisanship	-0.519** (0.160)	-0.600*** (0.150)	-0.625*** (0.155)
Log GDP per capita	-0.073 (0.118)	-0.069 (0.117)	-0.075 (0.114)
Unemployment	-0.003 (0.006)	0.001 (0.005)	-0.004 (0.006)
HDI	-2.932** (0.918)	-2.759** (0.903)	-2.726** (0.887)
Time FE	✓	✓	✓
Random Intercept	✓	✓	✓
Random Slope	✓	✓	✓
N	164	164	164

Notes: Standard errors in parentheses. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$. Models include year dummies, a random intercept for countries, and a random slope for negative voting and are weighted by election-year sample size. The reference category is two-party systems.

Figure A.1 Predicted incongruence by levels of negative voting across two-party and multiparty systems.



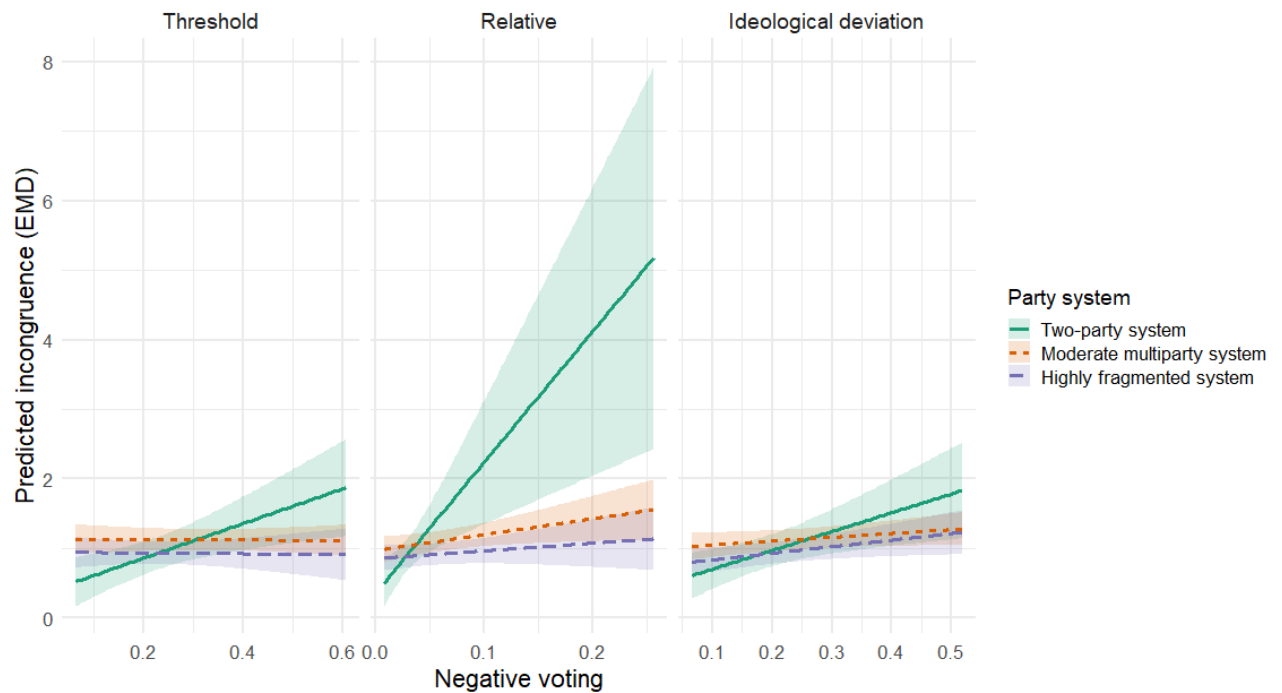
Note: The figure displays predicted values from the random intercept and slope interaction models across three operationalizations of negative voting. Each panel shows the average estimated relationship between negative voting and incongruence (EMD) by party system type, when holding all control variables constant. Around each linear regression line is a 95% confidence interval.

Table A.11 RE interaction models with three-category ENEP

Incongruence (EMD)			
Models	Threshold	Relative	Ideological threshold
Negative voting	2.494** (0.870)	18.807** (6.136)	2.710** (1.005)
Moderate multiparty system	0.768** (0.242)	0.617** (0.216)	0.567* (0.239)
Highly fragmented system	0.586* (0.242)	0.501* (0.215)	0.312 (0.245)
Negative voting × Moderate multiparty system	-2.516** (0.905)	-16.523** (6.168)	-2.162* (1.060)
Negative voting × Highly fragmented system	-2.551** (0.953)	-17.702** (6.212)	-1.762 (1.103)
Mean age	-0.014* (0.006)	-0.019** (0.006)	-0.013* (0.006)
Share weak partisanship	-0.379* (0.156)	-0.519*** (0.145)	-0.504** (0.153)
Log GDP per capita	-0.034 (0.112)	-0.035 (0.111)	-0.044 (0.110)
Unemployment	-0.002 (0.006)	0.000 (0.005)	-0.003 (0.005)
HDI	-3.254*** (0.878)	-3.118*** (0.867)	-3.053*** (0.857)
N	164	164	164

Notes: Standard errors in parentheses. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$. Models include year dummies, a random intercept for countries, and a random slope for negative voting, and are weighted by election-year sample size. . ENEP is divided into three categories: two-party systems ($ENEP < 3$), moderate multiparty systems ($ENEP \geq 3$ and ≥ 5) and highly fragmented systems ($ENEP \geq 5$). The reference category is two-party systems.

Figure A.1 Predicted incongruence by levels of negative voting across different levels of party system fragmentation



Note: The figure presents predicted values from the random intercept and slope interaction models across the three operationalizations of negative voting. Each panel shows the average estimated relationship between negative voting and incongruence (EMD) at different levels of party system fragmentation (ENEP), when holding all control variables constant. ENEP is divided into three categories: two-party systems ($ENEP < 3$), moderate multiparty systems ($ENEP \geq 3$ and ≥ 5) and highly fragmented systems ($ENEP \geq 5$). Around each linear regression line is a 95% confidence interval.